[533] (CONTINUATION)

On Factorials and Derivations.

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where the symbols $(\ )_k$ denote sums of products of the differences of an upper letter and a lower letter, θ factors in each product; and the several sums being formed as follows:

For $\begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $\alpha : \begin{array}{ll} \beta & a \\ \gamma & b \\ & c \\ & d \\ & e; \end{array}$

the expression is

$$(x-a)+(\beta-b)+(\gamma-c)+(\delta-d)+(\varepsilon-e)$$

For $\begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $\begin{array}{ll} \alpha,\ \alpha, & b \\ \alpha,\ \beta, & c \\ \alpha,\ \gamma, & d \\ \beta,\ \beta, & e \\ \beta,\ \gamma, & \\ \gamma,\ \gamma, & \end{array}$

viz. the expression is

$$(x-a)(x-b)+(x-a)(\beta-c)+\cdots+(\gamma-d)(\gamma-e)$$

For $\begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $\begin{array}{ll} \alpha,\ \alpha,\ \alpha, & c \\ \alpha,\ \alpha,\ \beta, & d \\ \alpha,\ \alpha,\ \gamma, & e \\ \alpha,\ \beta,\ \beta, & \\ \alpha,\ \beta,\ \gamma, & \\ \alpha,\ \gamma,\ \gamma, & \\ \beta,\ \beta,\ \beta, & \\ \beta,\ \beta,\ \gamma, & \\ \beta,\ \gamma,\ \gamma, & \\ \gamma,\ \gamma,\ \gamma, & \end{array}$

viz. the expression is

$$(x-a)(x-b)(x-c)+\cdots+(\gamma-c)(\gamma-d)(\gamma-e).$$

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For $\begin{pmatrix} \alpha,\ \beta \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $a,\ a,\ a,\ a,\quad a,\ b,\ c,\ d.$

viz. the expression is

$$(x-a)(x-b)(x-c)(x-d)+\cdots+(\beta-b)(\beta-c)(\beta-d)(\beta-e).$$

Finally for $\begin{pmatrix} \alpha \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, write $a,\ a,\ a,\ a,\ a,\quad a,\ b,\ c,\ d,\ e.$

viz. the expression is

$$(x-a)(x-b)(x-c)(x-d)(x-e),$$

which explains the law of the formation of the several coefficients. It is to be observed that in forming the development of any symbol, for instance $\begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}$, the first column contains the homogeneous products, θ together, of α, β, γ; the second column the combinations (that is, combinations without repetitions) θ together of a, b, c, d, e; the top line is a, a, c; and to form the subsequent lines we must for any advance a into β, &c. of a Greek letter make the like advance a into b, b into c, c into d, of the corresponding latin letter.

Two particular cases of the theorem may be noticed; if the latin letters all vanish, we have, for example,

$$\begin{aligned} \text{if } a = {} & (x-\alpha)(x-\beta)(x-\gamma)(x-\delta)(x-\varepsilon), \\ & + (b-a)(b-\beta)(b-\gamma)(b-\delta)\cdot H_1(\alpha,\ \beta,\ \gamma,\ \delta,\ \varepsilon) \\ & + (c-a)(c-\beta)(c-\gamma)\cdot H_2(\alpha,\ \beta,\ \gamma,\ \delta,\ \varepsilon) \\ & + (d-a)(d-\beta)\cdot H_3(\alpha,\ \beta) \\ & + (e-a)\cdot H_4(\alpha) \\ & + 1\cdot H_5(\alpha) \end{aligned}$$

where the symbols H denote the sum of the homogeneous products of the annexed letters, taken together according to the suffix number: the last coefficient $H_5(\alpha)$ is of course $=\alpha^5$. And if the greek letters all vanish, then we have in like manner

$$\begin{aligned} (b-a)(c-a)(d-a)(e-a) = {} & b^4 \\ & - b^3(a+b+c+d+e) \\ & + b^2(ab+\cdots+de) \\ & - b(abc+\cdots+cde) \\ & + abcde. \end{aligned}$$

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where the symbols C denote the combinations of the annexed letters taken together according to the suffix number; the last coefficient $C_3(a,\ b,\ c,\ d,\ e)$ is of course $= abcde$. This is the ordinary theorem giving the expression of an equation in terms of its roots.

Confining the two theorems, if in the first theorem we express the products $(x-a)(x-\beta)(x-\gamma)(x-\delta)\cdots+a$, &c. in powers of X by means of the second theorem; or if in the second theorem we express the powers b^4, b^3, &c. in terms of the products $(b-a)(b-b)(b-c)+(b-c)(b-d)$, &c. by means of the first theorem; then in either case we obtain certain identical relations connecting the C, H of $(\alpha,\ \beta,\ \ldots)$ or of $(a,\ b,\ \ldots)$.

I have mentioned the factorial notation

$$[n]^r = n(n-1)\cdots(n-r+1),$$

where r is a positive integer; a consequence of this is

$$[n]^{r+s} = [n]^r[n-r]^s,$$

where r and s are positive integers; or as this may also be written

$$[n+r]^{r+s} = [n+r]^r[n]^s.$$

Assuming this to subsist for $s=0$, or a negative integer; first for $s=0$, we have $[n+r]^r = [n+r]^r[n]^0$; that is, $[n]^0=1$, and we have for $s=-r$, we have $1=[n+r]^r[n]^{-r}$, that is,

$$[n]^{-r} = \frac{1}{[n+r]^r},$$

and in particular, $r=1,2,\&c.$, we have

$$[n]^{-1} = \frac{1}{n+1},$$

$$[n]^{-2} = \frac{1}{(n+1)(n+2)},$$

&c., which explains the extension of the factorial notation to negative integer values of the index.

But the equation

$$[n]^{r+s} = [n]^r[n-r]^s$$

does not in any determinate manner lead to an extension of the factorial notation to fractional or other values of the index. In fact, assuming $[n]^a = \frac{\Pi a}{\Pi(n-a)}$ where Π is an arbitrary fractional symbol, the equation in question becomes

$$\frac{\Pi a}{\Pi(n-r-s)} = \frac{\Pi n}{\Pi(n-r)}\cdot\frac{\Pi(n-r)}{\Pi(n-r-s)};$$

viz. the original equation is identically satisfied, without any condition whatever being imposed upon the function Π, and on this account we have not, in the notion of the factorial with an integer index, *any sufficient basis for a theory of general differentiation.*

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A product

$$n(n-r)\cdots(n-(r-1)\nu)$$

can of course be expressed in the factorial notation, viz. it is

$$= \nu^r \left[\frac{n}{\nu}\right]^r,$$

and on this account it is not in general necessary to employ a notation such as $[n, \nu]^r$ to denote such a factorial wherein the difference of the successive factors instead of being $= -1$ is $= -\nu$; in particular cases where factorials of the kind in question are used, it may be convenient to employ such a notation. In particular it is some- times convenient to use the notation $[n, -1]^r$ or better $[n]^r$ to denote the product

$$n(n+1)\cdots(n+r-1),$$

where the successive factors instead of being diminished, are increased by unity. It may be noticed, that reverting to this last product the order of the factors, we find

$$[n]^r = [n+r-1]^r;$$

a somewhat similar formula, but employing only the ordinary factorial notation, is obtained from the equation

$$[n]^r = n(n-1)\cdots(n-r+1)$$

by first changing the sign of n and then reversing the order of the factors; viz. we have

$$[-n]^r = (-)^r[n+r-1]^r.$$

Reverting to the process used for the development of the expression $(\ )_k$ where there are two columns, the one of greek, the other of latin letters; it is to be remarked that although the order in which the successive lines are evolved is not material for the purpose of the theorem, yet that a certain definite order of evolution, has been made use of; thus in regard to $\begin{pmatrix} \alpha,\ \beta,\ \gamma \\ a,\ b,\ c,\ d,\ e \end{pmatrix}_3$, the column of greek letters, giving the homogeneous products of the second order in (α, β, γ), was

$$\begin{array}{l} a\,a \\ a\,\beta \\ a\,\gamma \\ \beta\,\beta \\ \beta\,\gamma \\ \gamma\,\gamma \end{array}$$

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this is evolved from the top term (α, α) by a process given implicitly in Arbogast's Calculus of Derivations, and which may be termed the *rule of the last and the last but one.* Let the direction "operate on any letter," be understood to mean that the letter in question is to be changed into that which immediately follows it, but in such wise that when the letter occurs more than once, e.g. as in α, α the operation affects only the letter in the right-hand place. Then operate on the α, α in regard to α, we obtain α, β; operate on this in regard to β, we obtain α, γ; and in regard to α, we obtain β, β. Again we operate on α, γ in regard to γ, and obtain a, δ; we do not operate on it in regard to α for the reason that α is not the letter immediately preceding γ. Operate on β, β in regard to β, we obtain β, γ. The next step, if the series extended to ε would be to operate on α, δ in regard to δ, obtaining α, ε; on α, δ. Passing then to the next term β, γ, we operate in regard to γ, obtaining β, δ, and since β is the letter immediately preceding γ, we also operate in regard to β, obtaining γ, γ. Similarly if ε were admissible, β, δ would give β, ε, but it in fact gives nothing; γ, γ gives γ, δ; thus if ε were admissible would give γ, ε and δ, δ, but it in fact gives only δ, δ, and, ε being inadmissible, the process is here concluded. The rule is, operate on the last but one letter, and when the last but one letter is that which, in alphabetical order, immediately precedes the last letter (but in this case only) operate on the last but one letter.

Taking another example, but with numbers instead of letters, and supposing the highest admissible number to be 5, then from 111 we derive as follows:

111	112	113	114	115	125	135	145	155	255	355	455	535
		122	123	124	134	144	225	245	345	445		
			222	133	224	225	244	355	444			
				223	233	234	334	344				
						333						

the original single column being here for greater convenience broken up into distinct columns; but the order of the terms, when the columns are taken one after the other in order, each being read from the top to the bottom, being the same as before; it will be noticed that the successive divisions are the partitions into 3 parts (no part exceeding 5) of the numbers 3, 4, ..., 15 respectively; the partitions being in each case obtained without repetition, and those of the same number being given, say in their numerical order (corresponding with the alphabetical order when letters are employed). It is necessary to show that the partitions will be obtained without repetitions; and that all the partitions will be obtained; for this purpose consider, for example, the partitions of 9; any one of these is either a partition 135 where the last number 5 is not a repeated number; and to this case there is a partition of 8, viz. 134, from which operating on the last we obtain 135; but there is no other partition of 8 which would give 135, the only such partition would be 125, but here, as 2 is not the number which immediately precedes 5, there is no operation on the last but one, and we do not from it obtain 135. Or else a partition of 9 is of the form 144

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where the last letter is repeated; there exists in this case no partition of 8, such that operating on the last we obtain from it 144, but there does exist a partition of 8, viz. 134, such that operating on the last but one we obtain from it 144. That is, the partition 135 must be obtained from 134, and only from 134, and the partition 144 must also be obtained from 134, and only from 134, and that without repetitions, the entire series of the partitions of 9; and so in general.

Translating the example into letters, but using for greater convenience α, &c. instead of a, α, &c. the process will be precisely the same; taking the letters to be α, β, γ, δ, ε, we have

$$\begin{array}{ccccccccccccc}
\alpha\alpha\alpha & \alpha\alpha\beta & \alpha\alpha\gamma & \alpha\alpha\delta & \alpha\alpha\varepsilon & \alpha\beta\varepsilon & \alpha\gamma\varepsilon & \alpha\delta\varepsilon & \alpha\varepsilon\varepsilon & \beta\varepsilon\varepsilon & \gamma\varepsilon\varepsilon & \delta\varepsilon\varepsilon & \varepsilon\varepsilon\varepsilon \\
 & & \alpha\beta\beta & \alpha\beta\gamma & \alpha\beta\delta & \alpha\gamma\delta & \alpha\delta\delta & \beta\beta\varepsilon & \beta\delta\varepsilon & \gamma\delta\varepsilon & \delta\delta\varepsilon & & \\
 & & & \beta\beta\beta & \alpha\gamma\gamma & \beta\beta\gamma & \beta\beta\delta & \beta\gamma\delta & \gamma\gamma\varepsilon & \delta\delta\delta & & & \\
 & & & & \beta\beta\beta & \beta\gamma\gamma & \beta\gamma\delta & \gamma\gamma\delta & \gamma\delta\delta & & & & \\
 & & & & & & \gamma\gamma\gamma & & & & & &
\end{array}$$

Attributing weights to the several letters, viz. to α, β, γ, δ, ε the weights 1, 2, 3, 4, 5 respectively, the several columns show the terms of the weights 3, 4, ... 15 respectively.

I have said that the foregoing rule is given implicitly in Arbogast's Calculus of Derivations; this calculus includes in fact a process for the expansion of a function

$$\phi(x + hx' + \nu x'' + kx''' + \&c.)$$

in powers of x, the expansion in question may be obtained by means of Taylor's theorem, viz.

$$\phi(x + hx' + \nu x'' + kx''' + \&c.)$$

$$\begin{aligned}
&= \phi x \\
&\quad + \frac{\phi' x}{1}(hx' + \nu x'' + kx''' + \&c.) \\
&\quad + \frac{\phi'' x}{1\cdot 2}(hx' + \nu x'' + kx''' + \&c.)^2 \\
&\quad + \frac{\phi''' x}{1\cdot 2\cdot 3}(hx' + \nu x'' + kx''' + \&c.)^3 \\
&\quad + \&c.
\end{aligned}$$

viz. expanding the several powers of the polynomial increment, and arranging in powers of x, this is

$$\begin{aligned}
&= \phi x \\
&\quad + x'\phi' x \\
&\quad + x''\left[\phi' x + x'\phi'' x\right] \\
&\quad + x'''\left[\phi' x + x' x''\phi'' x + \frac{1}{6}x'^3\phi''' x\right] \\
&\quad + x''''\left[\phi' x + (x' x''' + \tfrac{1}{2}x''^2)\phi'' x + \tfrac{1}{2}x'^2 x''\phi''' x + \tfrac{1}{24}x'^4\phi'''' x\right] \\
&\quad + \&c.
\end{aligned}$$

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but the object of the rule is to obtain this last-mentioned result directly, and under- standing "operate in regard to a letter" to mean differentiate with regard to this letter and integrate with regard to the next succeeding letter, then the coefficients of the successive powers of x are all obtained from the first coefficient ϕx, by operating thereon according to the rule of "the last and the last but one"; thus ϕx, operating on α gives $\phi' x$, this operating in regard to α gives $\phi' x$, and in regard to α gives $\phi'' x \cdot \frac{x'}{1}$; the term $\phi'' x \cdot x'$ is to be operated upon in regard to α only, and is given $\phi'' x \cdot x''$; the other terms $\phi'' x \cdot \frac{x'^2}{2}$ operated on in regard to α gives $\phi'' x \cdot \nu x$, and in regard to α gives $\phi'' x \cdot \frac{x'^2}{2}$, and so on. But attending only to the literal parts, the terms, for instance b', $b\nu$, νb, &c., which present themselves in the formula, are the homogeneous terms derived from ν^3, by the rule, as originally stated, with a view to the derivation of such terms.

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A "Smith's Prize" Paper(†); Solutions.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. V. *(1870), pp. 182–203.]*

1. Mention what forms of plane relation $\phi(x, b, c, \ldots) = 0$ between the roots of a plane equation will in general serve for the rational determination of the roots; explain the case of failure; and state what information as to the roots is furnished by a plane relation not of the form in question.

In the given relation $\phi(x, b, c, \ldots) = 0$ must be a wholly unsymmetrical function of the roots; that is, a function altered by any permutation whatever of the roots; or, what is the same thing, by any single interchange of two roots.

For this being so, if a, β, γ, δ, and the corresponding four values of a, b, c, $\ldots$ satisfied by writing therein $a = a$, $b = \beta$, $c = \gamma$, &c., it will in general be satisfied also by writing $a = \beta$, $b = a$, $c = \gamma$, &c.; but the equation must uniquely determine each of the roots a, b, $\ldots$, respectively, in a separate question which is not here considered.

The function $\phi(a, b, c, \ldots)$ may be of the proper form, and yet, for particular values a, β, γ, $\ldots$, be such that the given relation $\phi(a, b, c, \ldots) = 0$ is satisfied, not only for the single arrangement $a = a$, $b = \beta$, $c = \gamma$, &c., but for some other arrangement,

†Set by me for the Master of Trinity, Feb. 5, 1870.

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$a = \delta$, $b = \gamma$, $c = \beta$, ... or for more than one such other arrangement. (For instance, if the given relation be $a + b + c - 2c = 0$, and the roots are 3, 5, 7; the relation is satisfied by $a = 5$, $b = 3$, $c = 7$, and also by $a = 3$, $b = 7$, $c = 5$.) Here the given equation and relation do not completely determine each root, they only determine that a is $= a$ or $= \delta$ (or as the case may be $=$ some other one value); and similarly that b is $= \beta$ or $= \gamma$ (or as the case may be $=$ some other one value), and so for the other roots c, d, ...; and it then appears *a priori*, that in such a case each root is determined, not rationally, but by means of an equation, the order of which is equal to the number of the values of such root; we have here the case of failure of the general theorem.

When the given relation $\phi(a, b, c, \ldots) = 0$ is not of the required form; that is, when $\phi(a, b, c, \ldots)$ is a partially symmetrical function, there will be in general several arrangements of a, β, γ, ..., such that equating $\phi(a, b, c, \ldots) = 0$ will be satisfied; for each of these arrangements, the given relation $\phi(a, b, c, \ldots) = 0$ will be satisfied; and it follows that each of the roots a, b, c, ... is determined not rationally, but by means of an equation of a certain order (not always quadratic equation for the several values of each of the several arrangements, and as the same value, viz., that is, if it be satisfied, suppose by $a = \alpha$, $b = \beta$, $c = \gamma$, ..., it will also be satisfied by $a = \beta$, $b = \alpha$, $c = \gamma$, ..., but not in general in any other manner; each of the roots a, b has here either of the values a, β, and the two roots a, b in question will be given, not rationally, but by means of the same quadratic equation. And observe, moreover, that any other function $\psi(a, b, c, \ldots)$ of the same form as ϕ, that is, symmetrical in regard to the two roots a, b, will have for its two arrangements $a = \alpha$, $b = \beta$, and $a = \beta$, $b = a$, the same value, $c = \gamma$, ... acquire not two distinct values, but one and the same value, the value of $\psi(\alpha, \beta, c, \ldots)$ will be determined *rationally*; and so in general.

There is for the partially symmetrical function $\phi(a, b, c, \ldots)$ a case of failure similar to that which arises for the completely unsymmetrical function, viz. the particular values a, β, γ, ... may be such as to give more ways of satisfying the given relation $\phi(a, b, c, \ldots) = 0$, than there would be in general for partially symmetrical values of a, β, γ, ... and there is a corresponding elevation of the order of the equation for the determination of the roots a, b, c, ... or some of them.

2. If the roots of $(a, \beta, \gamma, \delta)$ of the equation

$$(a, b, c, d, e)(x, 1)^4 = 0$$

are no two of them equal; and if there exist unequal magnitudes θ, ϕ such that

$$(\theta + \alpha)^4 : (\theta + \beta)^4 : (\theta + \gamma)^4 : (\theta + \delta)^4 = (\phi + \alpha)^4 : (\phi + \beta)^4 : (\phi + \gamma)^4 : (\phi + \delta)^4;$$

show that the cubinvariant ace $-ad^2 - b^2e - c^3 + 2bcd$ is $= 0$; and find the values of θ, ϕ.

We have

$$\left(\frac{\theta + \alpha}{\phi + \alpha}\right)^4 = \left(\frac{\theta + \beta}{\phi + \beta}\right)^4 = \left(\frac{\theta + \gamma}{\phi + \gamma}\right)^4 = \left(\frac{\theta + \delta}{\phi + \delta}\right)^4;$$

and we cannot have any two of the fourth roots, say $\dfrac{\theta + \alpha}{\phi + \alpha}$ and $\dfrac{\theta + \beta}{\phi + \beta}$ equal to each other; for this would imply $(\theta - \phi)(a - \beta) = 0$, that is, $\theta = \phi$ or else $\alpha = \beta$.

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Hence assuming $\dfrac{\theta+\alpha}{\phi+\alpha}=a_1$, $\dfrac{\theta+\beta}{\phi+\beta}=a_2$, $\dfrac{\theta+\gamma}{\phi+\gamma}=a_3$, $\dfrac{\theta+\delta}{\phi+\delta}=a_4$, viz. this is one of three notations of equations; the other two may be obtained from a_4 by writing a_3, a_2, a_1, or by interchanging some of the roots, a_1, a_2, a_3, a_4 are in fact the imaginary fourth roots of unity, and the corresponding four values of $(\alpha,\ \beta,\ \gamma,\ \delta)$ is therefore equal to four values a, b, c, d. We have therefore

$$\frac{\theta+\alpha}{\phi+\alpha}=-1,\quad \frac{\theta+\beta}{\phi+\beta}=+1$$

$$\frac{\theta+\gamma}{\phi+\gamma}=-\sqrt{-1},\quad \frac{\theta+\delta}{\phi+\delta}=+\sqrt{-1}$$

which are different cases, related to each other; the mathematic ratio of $(\alpha,\ \beta,\ \gamma,\ \delta)$ is therefore equal to that of $(1,\ -1,\ \sqrt{-1},\ -\sqrt{-1})$;

That is,

$$(\alpha-\gamma)(\beta-\delta)=(-1-\sqrt{-1})(1+\sqrt{-1})$$

or, what is the same thing,

$$2(\alpha\beta+\gamma\delta)=(\alpha+\beta)(\gamma+\delta);$$

viz. we have this relation, or one of the like relations

$$2(\alpha\gamma+\beta\delta)=(\alpha+\gamma)(\beta+\delta),$$

$$2(\alpha\delta+\beta\gamma)=(\alpha+\delta)(\beta+\gamma),$$

that is, the product of the three functions $2(\alpha\beta+\gamma\delta)-(\alpha+\beta)(\gamma+\delta)$, &c. is $=0$.

But the product of these is the cubinvariant of the quartic function $(a,\ b,\ c,\ d,\ e)$. viz. we have this relation, or one of the two following: $J=0$,

$$J=ace-ad^2-b^2e-c^3+2bcd,$$

which is evidently a transcendental curve; it may however be shown that, if

$$x^4+y^4=1$$

and

$$x(1+x^2)+y(1+y^2)-2xy(1+x+y)(1+x+y)=0,$$

that is, if we have the equation $\dfrac{\theta+\alpha}{\phi+\alpha}=-1$, etc. then there is a relation in y which gives $J=0$. By the consideration of the form $-r^4-r^4-r^4-r^4+r^4+r^4+r^4+r^4=0$, which is the form $J=0$ as it may by any means easy to prove that this is so. The question depends on the form of the curve defined by the equation

$$4y^4-4z(z+by^2+cy^2)=0.$$

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That is,

$$2b\phi + 2a\theta - (\theta + \phi)(a + \beta) = 0,$$

or, what is the same thing,

$$2b\phi + 2a\theta - (\theta + \phi)(\gamma + \delta) = 0;$$

viz. we have thus the values of $\theta\phi$, $\theta + \phi$ each given in terms of (a, β) corresponding to the relation $2(a\beta + \gamma\delta) = (a + \beta)(\gamma + \delta)$; and there are by cyclically permuting X, Y, Z as before, we have the values of $\theta\phi$, $\theta + \phi$ corresponding to the other two forms respectively of the relation between the roots.

3. If on a plane A, B, C, D are fixed points and P a variable point, find the locus relations

$$\alpha \cdot PAB + \beta \cdot PBC + \gamma \cdot PCD + \delta \cdot PDA = 0$$

which connects the areas of the triangles PAB, PBC, PCD, PDA.

Taking (x, y), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4) for the coordinates of P, A, B, C, D respectively, we have

$$PAB = \frac{1}{2}\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}, \quad \text{etc.,} \quad \text{suppose,}$$

$$PBC = 023, \ \&c.$$

(where the values of these numerical determinants do depend the signs of the several triangles). The identical equation is

$$\alpha \cdot 012 + \beta \cdot 023 + \gamma \cdot 034 + \delta \cdot 041 = 0$$

(that such an equation exists appears at once by the consideration that α, β, γ, δ can be determined so that the coefficients of x, y, and the constant term shall severally vanish); and in order actually to find the values we may take P coincident with the points A, B, C, D respectively. We have thus

$$\begin{aligned}
\beta \cdot 123 + \gamma \cdot 134 + \delta \cdot 141 &= 0,\\
\gamma \cdot 234 + \delta \cdot 241 + \alpha \cdot 212 &= 0,\\
\delta \cdot 341 + \alpha \cdot 312 + \beta \cdot 323 &= 0,\\
\alpha \cdot 412 + \beta \cdot 423 + \gamma \cdot 434 &= 0,
\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}
\beta \cdot 123 + \gamma \cdot 134 + \delta \cdot 141 &= 0,\\
\gamma \cdot 234 + \delta \cdot 241 + \alpha \cdot 212 &= 0,\\
\delta \cdot 341 + \alpha \cdot 312 + \beta \cdot 323 &= 0,\\
\alpha \cdot 412 + \beta \cdot 423 + \gamma \cdot 434 &= 0.
\end{aligned}$$

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and these are at once seen to give

$$\alpha : \beta : \gamma : \delta = 234 : 341 : 412 : 123 = -341 : 412 : -123 : 234,$$

so that, the required identical relation is

$$234 \cdot 012 - 341 \cdot 023 + 412 \cdot 034 - 123 \cdot 041 = 0,$$

or, what is the same thing,

$$PAB \cdot BCD - PBC \cdot CDA + PCD \cdot DAB - PDA \cdot ABC = 0,$$

which signifies that this signs being attended to the signs of the triangles PAB, PCD, PDA, BCD, CDA, DAB, ABC, are taken with respect to the hexagon of the point.

4. Find at any point of a plane curve the angle between the normal and the line drawn from the point to the centre of any one of the two circles of curvature.

Excursus whether a like question applies to a point on a surface and the indicatrix section at such point.

Taking the origin at the point on the curve, and considering as the axis of x the tangent and that of y with the normal, the equation of the curve is

$$y = bx^2 + cx^3 + \dots,$$

and if, considering x as small quantity of the first order, and therefore y as a small quantity of the second order, we regard y as given, and find the two values x_1, x_2 such of the order $\sqrt{y}$, which satisfy the equation, then, as will appear, $x_1 + x_2$, is a small quantity of the order $\sqrt{y}$, and consequently $\frac{1}{2}(x_1 + x_2)$ is of the order y, that is if ϕ be the required angle, then obviously $\tan\phi = \frac{1}{2y}(x_1 + x_2)$.

Taking the origin at the point on the curve, the original equation is the usual approximation $bx^2 = y$, $cx^3 = y\left(1 - \frac{x}{b}\right)$, whence $x = \sqrt{\frac{y}{b}}\left(1 - \frac{c}{2b}\frac{x}{b}\right)$, $cx^3 = \frac{1}{2}y$, &c., so we have

$$x_1 = \sqrt{\frac{y}{b}} - \frac{cy}{2b^2},$$

$$x_2 = -\sqrt{\frac{y}{b}} - \frac{cy}{2b^2},$$

and thence

$$\tfrac{1}{2}(x_1 + x_2) = -\frac{cy}{2b^2},$$

which gives the value of the angle, viz. obviously $\tan\phi = -\frac{c}{b^2}$; which gives the value of the angle ϕ; it would be easy to express b, c in terms of the differential coefficients

$$\frac{dx}{dy}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}.$$

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It would at first sight appear that a like question might be asked as to a surface; viz. that it might be proposed to determine the angle between the normal and a line drawn from the point to the centre of the indicatrix conic. But this is not so; in fact, taking the origin at a point on the surface, the axis of z being in the tangent plane, and that of z coinciding with the normal; then for the third order we have

$$z = (A,\, B,\, C \breve{\,} x,\, y)^2 + (a,\, b,\, c,\, d \breve{\,} x,\, y)^3;$$

but here, regarding z as a given constant, if we take account of the terms of the third order, the section is not a conic but a cubic; and it has not in general any centre; and if (as in the ordinary theory) we neglect the terms of the third order, thus obtaining an indicatrix conic, the centre of this conic lies on the normal, and there is no angle corresponding to the angle ϕ of the plane problem.

The only case where there is no objection is when the cubic terms $(a,\, b,\, c,\, d \breve{\,} x,\, y)^3$ contain as a factor the quadric terms $(A,\, B,\, C \breve{\,} x,\, y)^2$ (one relation between the coefficients a, b, c, d of the cubic function depending on this relation), and in such case the indicatrix section corresponds to the surface plane $z =$ (a coincident section: in this case may be defined as a section by a plane parallel to the tangent plane, but at any rate close to it as a coincident plane. To complete the analogy, (2) the two functions depend in any manner whatever on one and the same set of constants.

5. If V, V' are binary functions of the form $(\alpha,\, b,\, \ldots \breve{\,} x,\, 1)^n$ with arbitrary coefficients, and if the equation $U = 0$, $V = 0$ have a common root, show how this can be determined in terms of the derived functions of the Resultant in regard to the coefficients of either function.

Show what results in regard to the common root can be obtained when the coefficients are not all of them arbitrary but (1) each or either of the functions depends in any manner whatever on a set of arbitrary coefficients not entering into the other function, (2) the two functions depend in any manner whatever on one and the same set of arbitrary coefficients.

Here is the theory analytical at once, *instead of the two equations, there is a single equation* $U = 0$ *having a double root.*

Suppose

$$U = (a,\, b,\, \ldots \breve{\,} x,\, 1)^m, \quad \left(= ax^m + \frac{m}{1} bx^{m-1} + \&c.\right),$$

$$V = (a,\, b,\, \ldots \breve{\,} x,\, 1)^n, \quad \left(= ax^n + \frac{n}{1} bx^{n-1} + \&c.\right).$$

Then if R is the resultant, the equation $R = 0$ is the relation which must exist between the coefficients in order that the two functions U and V shall have a common root. Moreover, if we assume that U and V may have a common root $x = \alpha$, then we have $U_\alpha = 0$, $V_\alpha = 0$; the coefficients of U_α, V_α being the same as those of U, V, and $x = \alpha$. We have then identically $U_\alpha = 0$, $V_\alpha = 0$, $R = 0$; that is, these equations have to be infinitesimally varied in such manner that U, V have still a common root $x = \alpha$, we have the equation $U = 0$, $V = 0$ subsist when a, b, …, the coefficients of U, and a, b, … those of V undergoes, the given relation must be a common root, a this implies between U_α, V_α, R_α &c.

$$da \frac{dR}{da} + db \frac{dR}{db} + \&c. = 0.$$

But the common factor $x = \alpha y$ is a factor of the unaltered equation $U = 0$, and the values of $(x,\, y)$ are thus unaltered, viz. the equation $U = 0$ is satisfied with the original values of $(x,\, y)$; that is, changing the α as a small quantity of the first order, we have

$$x^m da + mx^{m-1} y\, db + \&c. = 0,$$

or, what is the same thing,

$$x^m\sigma + mx^{m-1}y\sigma' + \&c. = 0$$

an equation which must agree with the former one, that is, we have

$$x^m : mx^{m-1}y : \cdots = \frac{dR}{da} : \frac{dR}{db} : \cdots ,$$

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5. Show that a cubic surface has at most four conic points; and a quartic surface at most sixteen conical points.

If a cubic surface has two conical points, then the line joining these has with the surface two intersections at each of the conical points, and therefore lies wholly in the surface. Hence, for a cubic surface with three conical points A, B, C, the lines AB, BC, CA lie wholly in the surface, and three more lines, viz. ones formed by the lines through three conical points A, B, C, D; the line AB, &c. wholly in the surface, and these lines form among them the complete section of the surface by a plane wholly through the conic; this is the case of complete section of the surface, and the cubic surface is in fact equal to the cubic surface as is well known the cubic surface contains four points E, and the line joining E with any of the four conical points has, by the line, three points of contact with the surface, and is therefore not a tangent line. Hence the section of a cubic surface by a plane through three conical points A, B, C, would be the triangle ABC taken twice, and a line which is the locus of points E; but each of the points E, E_1 in particular, would have to be conical points, and the cubic surface would have 5, that is, more than four conical points, which is impossible.

6. *Find the differential equation of the parallel surfaces of an ellipsoid.*

Let $(x,\, y,\, z)$ be the coordinates of a point on the ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}-1=0$; $(X,\, Y,\, Z)$ the coordinates of a point on the normal at a distance $= k$ from the foot- mentioned point. We have

$$\frac{X-x}{\frac{x}{a^2}}=\frac{Y-y}{\frac{y}{b^2}}=\frac{Z-z}{\frac{z}{c^2}}=\mu \quad \text{suppose;}$$

that is,

$$X=x\left(1+\frac{\mu}{a^2}\right),\quad Y=y\left(1+\frac{\mu}{b^2}\right),\quad Z=z\left(1+\frac{\mu}{c^2}\right);$$

and thence

$$k^2=\mu^2\left\{\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4}\right\}.$$

Moreover

$$\frac{X^2}{(a^2+\mu)^2}+\frac{Y^2}{(b^2+\mu)^2}+\frac{Z^2}{(c^2+\mu)^2}=\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4};$$

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substituting these values in the equation of the ellipsoid, we have

$$1 = \frac{a^2X^2}{(a^2+\mu)^2} + \frac{b^2Y^2}{(b^2+\mu)^2} + \frac{c^2Z^2}{(c^2+\mu)^2},$$

which determines μ as a function of X, Y, Z. The tangent plane of the ellipsoid at the point (x, y, z) and of the parallel surface at the point (X, Y, Z) are parallel to each other (for what is the same thing, the parallel surface cuts at right angles on the normal of the ellipsoid); we have therefore

$$\frac{X}{a^2}dX + \frac{Y}{b^2}dY + \frac{Z}{c^2}dZ = 0,$$

or substituting for x, y, z their values, we have

$$\frac{XdX}{a^2+\mu} + \frac{YdY}{b^2+\mu} + \frac{ZdZ}{c^2+\mu} = 0,$$

where μ denotes as above a function of (X, Y, Z) given by the equation

$$1 = \frac{a^2X^2}{(a^2+\mu)^2} + \frac{b^2Y^2}{(b^2+\mu)^2} + \frac{c^2Z^2}{(c^2+\mu)^2}.$$

We have thus the differential equation of the parallel surface. It may be remarked, that the integral equation (involving k as the constant of integration), is found by the elimination of μ, x, y, z from the foregoing equations

$$X - x = \mu\frac{x}{a^2}, \quad Y - y = \mu\frac{y}{b^2}, \quad Z - z = \mu\frac{z}{c^2},$$

or, what is the same thing, by the elimination of μ from the equations

$$\frac{X^2}{(a^2+\mu)^2} + \frac{Y^2}{(b^2+\mu)^2} + \frac{Z^2}{(c^2+\mu)^2} = \frac{1}{\mu^2}k^2,$$

$$\frac{a^2X^2}{(a^2+\mu)^2} + \frac{b^2Y^2}{(b^2+\mu)^2} + \frac{c^2Z^2}{(c^2+\mu)^2} = 1;$$

these may be replaced by

$$\frac{X^2}{(a^2+\mu)^2} + \frac{Y^2}{(b^2+\mu)^2} + \frac{Z^2}{(c^2+\mu)^2} = \frac{k^2}{\mu^2},$$

$$\frac{X^2}{a^2+\mu} + \frac{Y^2}{b^2+\mu} + \frac{Z^2}{c^2+\mu} - 1 = 0,$$

or, since here the second equation is the derived equation of the first in regard to the parameter μ, the parallel surface is the envelope of the quadric surface

$$p(\mu) + p(\mu) + p(\mu) - 1 = 0, \quad p(\mu) = \frac{X^2}{a^2+\mu} + \frac{Y^2}{b^2+\mu} + \frac{Z^2}{c^2+\mu}.$$

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7. *Explain wherein consists the peculiarity of the following problem, and solve it by geometrical considerations:—*

Determine the knot circle inclosing three given points.

The peculiarity of the problem is that the variable parameters upon which the circle depends, viz. a, β the coordinates of the centre and ρ its radius, are not all subject to any equations, but only to the inequalities

$$b^2 + (a - x_1)^2 \leq \rho^2, \quad b^2 + (a - x_2)^2 \leq \rho^2, \quad b^2 + (a - x_3)^2 \leq \rho^2,$$

(α_1, β_1), (α_2, β_2), (α_3, β_3), the coordinates of the given points, and the $\leq$ indicates that the points are within or on the circumference of the circle, and the equation $\rho =$ minimum. The parallel surface inclosing within or on the circumference of the circle, and the points A, B, C separately. Then for the values of a, β, the second sides of the (a, β, ρ) the variables of the given points, with the condition that the points A, B, C are within or on the circumference; and then ρ minimum.

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But if a be the acoustic anomaly, then

$$x = a\sin u, \quad y = b\sin u, \quad a = \sqrt{1-e^2}\sin u,$$

and the equations become

$$-\sin u\frac{d^2u}{dt^2} + \cos u\left(\frac{du}{dt}\right)^2 - \frac{\sin u}{a^2} + \frac{\cos u}{a^2(1-e\sin u)^2}\cos u = 0,$$

$$+\cos u\frac{d^2u}{dt^2} + \sin u\left(\frac{du}{dt}\right)^2 - \frac{\cos u}{a^2(1-e\sin u)^2}\sin u = 0,$$

and multiplying by $-\cos u$, $-\sin u$, respectively, and adding, we have

$$\left(\frac{du}{dt}\right)^2 = \frac{1}{a^2(1-e\sin u)^2 - e^2a^2\sin^2 u}\cdot\frac{1}{1+e\sin u},$$

which is the required differential equation.

9. Show that the attraction of an indefinitely thin double-convex lens at a point at the centre of one of its faces is equal to that of the infinite plate included between the tangent plane at the point and the parallel tangent plane of the other face of the lens.

The figure represents the upper half only of the lens, but in speaking of any portion thereof, such as PRQ, we include the symmetrically situate portion of the under-half of the lens.

Let a, $= PQ$, be the thickness of the lens, LPQ a circle of which angle is ultimately $= \frac{a}{a}$. Then it is at once seen that the attraction of the cone LPQ is $=$ const $= 2\pi a$; and from this it follows that the attraction of the infinite plate is $= 2\pi a$. The attraction of the whole infinite plane (resp. the cone LPQ) is $= 2\pi a$, which is indefinitely small in regard to $2\pi a$; and, *a fortiori*, the attraction of the portion MPR of the lens is indefinitely small in regard to $2\pi a$. We have then only to show that the attraction of the mass LRQ is indefinitely small in regard to $2\pi a$; the mass being, in fact, indefinitely thin in the direction of the lens, the attraction of the cone MRQ will not therefore ultimately be $= 2\pi a$.

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Let the position of an element of the solid in question be determined by r the distance from P, θ the inclination of r to the axis PQ and ϕ the azimuth in regard to any fixed plane through the axis; then $dm = r^2 \sin\theta\, dr\, d\theta\, d\phi$, the integral in regard to ϕ having been taken from 0 to 2π, and the attraction in the direction in the figure: PQ is $= \iint a \sin\theta \cos\theta\, dr\, d\theta - \frac{1}{2}\iint \sin 2\theta\, dr\, d\theta$, the integral in regard to ϕ having been taken from 0 to π; and the attraction in the direction of PQ from any point $A =$ a constant when $\phi = ar$, and that for $\phi = a$, the limit is the small circle. The plane PRQ, $(Z = 0)$ the equation will be $\phi =$ const.

Taking the radius of the small circle to be $= \rho$, then for a fixed value of ϕ the value of r extends from 0 to $\frac{2\rho\cos\theta}{1-e\sin\theta}$, while the integral in regard to ϕ is to be taken from $\phi = 0$ for the value of r for which $\theta = \tan^{-1}\frac{a}{\rho}$ to $\phi = \frac{\pi}{2}$. We have then

$$\frac{1}{2}a^2 \int \sin 2\theta\, d\phi = (1+e)\sin\theta\cos\theta\,(1+e^2\sin^2\theta)\, d\theta,$$

and the integral is to be taken from $\theta = 0$ to $\theta = \tan^{-1}\frac{a}{\rho}$, viz. this is

$$\int \sin\theta\cos\theta\,(1-e\sin\theta)^2\,(1+e^2\sin^2\theta)^{-1}\, d\theta - a\, a^2\, a^{-2}\, d\theta,$$

so that taking this between the limits in question, we have

$$\begin{aligned}\frac{1}{2}a^2 \int \sin 2\theta\, d\phi &= (1+e)\sin^2\theta - \frac{1}{6}(1+e)^2(1-e\sin\theta)^3\\ &-(1+e)\frac{1}{4}\left[1-\frac{1}{(1+e)^2}\right] - \frac{1}{2}(1+e^2)\left[1-\frac{1}{1+e}\right]\\ &+\frac{1}{2}(1-2e^2)\frac{(1+e)^2+(1+e)}{1+e} - \frac{1}{3}(1-2e^2)\frac{(1+e)^3+1}{(1+e)^3}.\end{aligned}$$

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viz. a here an indefinitely small quantity of the order a^2; all the terms are therefore at least of the order a^2, and are to be neglected in comparison with $2\pi a$, the integral work shown we have $A = 0$ (that is, the attraction of the mass LRQ is indefinitely small in the standard sense (since $L = a$ to the doctrine in the form RPQ is mass equal in $2\pi a$), as the theorem is thus proved.

10. *Indicate in what manner the Lagrangian equations of motion*

$$\frac{d}{dt}\frac{dT}{dx'} - \frac{dT}{dx} + \frac{dV}{dx} = 0, \quad \&c.$$

lead to the equation

$$\frac{d^2x}{dt^2} + (U - R)\frac{dV}{dy} = 0, \&c.$$

for the motion of a solid body about a fixed point.

The expression of the vis viva theorem T is

$$T = \tfrac{1}{2}(Ap^2 + Bq^2 + Cr^2);$$

but this expression will not by itself lead to the equations of motion; we require to know also the expression of p, q, r in terms of certain coordinates λ, μ, ν, which determine the position of body in regard to axes fixed in space, and of the differential coefficients λ', μ', ν' of these coordinates in regard to the time; each of the quantities p, q, r will be a linear function of λ', μ', ν', ($p = a\lambda' + b\mu' + c\nu', \&c.$), containing in any manner whatever the coordinates λ, μ, ν, and the equations of motion will be

$$A\frac{dp}{dt}\frac{dq}{d\lambda} + B\frac{dq}{dt}\frac{dr}{d\mu} + C\frac{dr}{dt}\frac{dp}{d\nu} = 0, \quad \&c.$$

where $\frac{dp}{d\lambda}$, $\frac{dq}{d\mu}$, $\frac{dr}{d\nu}$ are each independent of λ', μ', ν'; hence, in the equation, the only terms containing the differential coefficients of λ', μ', ν' are the terms

$$A\frac{dp}{dt}\frac{dq}{d\lambda'} + B\frac{dq}{dt}\frac{dr}{d\mu'} + C\frac{dr}{dt}\frac{dp}{d\nu'};$$

of $\frac{dT}{dx'} - \frac{dT}{dx}$; and hence, assuming that the equations of motion on the known equations $A\frac{dp}{dt} + (C - B)qr = 0$, it appears that the equation $A\frac{d^2x}{dt^2} + (U - R)\frac{dV}{dx} = 0$ may be put

$$\frac{d}{dt}\left[A\frac{dp}{dt}\cdot\frac{dq}{d\lambda'} + B\frac{dq}{dt}\cdot\frac{dr}{d\mu'} + (C - B)qr\frac{dp}{d\nu'}\right] = 0;$$

$$\frac{d}{dt}\left[A\frac{dp}{dt}\cdot\frac{dq}{d\mu'} + B\frac{dq}{dt}\cdot\frac{dr}{d\nu'} + (A - C)rp\frac{dq}{d\lambda'}\right] = 0;$$

$$\frac{d}{dt}\left[A\frac{dp}{dt}\cdot\frac{dq}{d\nu'} + B\frac{dq}{dt}\cdot\frac{dr}{d\lambda'} + (B - A)pq\frac{dr}{d\mu'}\right] = 0;$$

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$\frac{dp}{d\lambda'}$, $\frac{dq}{d\mu'}$, $\frac{dr}{d\nu'}$, &c. in not $= 0$; and the three equations are therefore equivalent to (as they should be) to the equations

$$A\frac{dp}{dt} + (C - B)qr = 0, \quad \&c.$$

What precedes is a complete answer to the question, but in regard to the actual expression of p, q, r, it may be remarked, that these quantities may be expressed very symmetrically in terms of the quantities

$$\lambda, \mu, \nu = \tan\frac{1}{2}\theta\,\cos\phi, \tan\frac{1}{2}\theta\,\sin\phi, \tan\frac{1}{2}\theta,$$

which determine the position of the principal axes in regard to the axes fixed in space, by means of the angles of position (azim. ϕ and the resultant axis, and the notation of about this axis is θ; writing $a^2 + b^2 + c^2 = k^2$, we have, the equations

$$ap = (1 + b^2)\lambda' - (1 + c^2)\mu' + (\nu - h)\nu',$$

$$aq = (b + c)\lambda' - h^2\mu' - (\mu + g)\nu',$$

$$ar = (n - g)\lambda' + h\mu' + h^2\nu';$$

and the above result may be verified *a posteriori* without any difficulty. See Crelle's *Math. Jour.*, vol. XI. (1843), [6], p. 224, [Coll. Math. Papers, vol. I. p. 33].

11. *Find in the Hamiltonian form,*

$$\frac{dx}{dt} = \frac{dH}{dp}, \quad \frac{dp}{dt} = -\frac{dH}{dx}, \quad \&c.$$

the equations for the motion of a particle acted on by a central force.

Taking as coordinates r the radius vector, ω the longitude, y the latitude, the equation of the vis viva function is

$$T = \frac{1}{2}(r'^2 + r^2\cos^2 y\,\omega'^2 + r^2y'^2),$$

hence

$$\frac{dT}{dr'} = r' = a \quad \text{suppose},$$

$$\frac{dT}{d\omega'} = r^2\cos^2 y\,\omega' = \nu \quad ,$$

$$\frac{dT}{dy'} = r^2y' = \mu \quad ,$$

and the expression of T in terms of r, a, y and the new coordinates a, ν, μ is

$$T = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right);$$

whence writing

$$H = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right) + V,$$

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the equations are

$$\frac{dH}{dr} = a, \quad \frac{dH}{d\omega} = \nu, \quad \frac{dH}{dy} = \mu,$$

$$-\frac{dH}{dr} = a, \quad -\frac{dH}{d\omega} = \nu, \quad -\frac{dH}{dy} = \mu.$$

We have

$$\frac{dH}{dr} = -\frac{\nu^2}{r^3} - \frac{\mu^2}{r^3\cos^2 y} + \frac{dV}{dr},$$

$$\frac{dH}{dy} = \frac{\mu^2}{r^2\cos^2 y}\tan y + \frac{dV}{dy} \quad \frac{dH}{d\omega} = \frac{dV}{d\omega} = \frac{r^2\cos^2 y\,\omega'}{r^2\cos^2 y};$$

and substituting these values, the equations of motion present themselves in a system of the first order between r, a, y, μ, ν, and it is well known that the case of a central force corresponds to that of a function V where the function V contains r only.

12. *An ordered polygon of $(n+1)$ vertices is constructed as follows:— viz. the several vertices are taken at random of the unknown a. In each, considering as the largest order a the size of the whole polygon is very small; and may be regarded as being $= 1$, and the area between the ordinates corresponding to the several values of being $= 1$, are like ordinate. Find the probability of the unknown polygon being in the form of a convex polygon, and the corresponding probabilities respective ordinate.*

It is somewhat more convenient to take account also of the abscissa $= 1, \ldots = u$, thereby obtaining a polygon of $2n+1$ vertices, corresponding to the abscissa 1, in the largest, and the position of the polygon being on either side of the abscissa $= 1$, that is, the polygon is fixed at the abscissa $= a$, ever this, a being any given number, the probability of a number of heads between $n = a$ and $a + a$ increases with the number a being fixed and equally probable, all the probability of the form in question being so probably, and so on; the probability of the position of these large numbers of vertices: viz. the chance that the polygon has only a number of vertices equal to that of the first a of the second period, that is, the probability of a vertex polygon between the ABC, hence the two periods have its sum.

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The conclusion in regard to the areas of the polygons is that, taking A any given value whatever, however large, the ratio (so being of course $> A$) which the area between the ordinates of the abscissa $a{+}1, \ldots = a{+}n$ bears to the area of the whole polygon (or to unity) continually decreases as n increases, and ultimately vanishes; but con- trariwise, taking a way given function whatever, however small, the area which the respectively bears to the area of the whole polygon, however small, becomes ultimately $= 1$.

13. *Show that for the quartic surface $aU^2 + 2bUV + cV^2 + 2dW + e = 0$, where of the surface is a quartic surface through six points particular values* $(U,\ V,\ W,$ &c.)*; then the equation of the general quartic surface through the six points will be of the form*

$$(U \cdot V, W, \cdot) \cdot \alpha U + \beta V + \gamma W + \cdots = 0,$$

$(U, V, W, \cdot)$ *the coordinates of the vertex, in regard to the unknown.*

Suppose $U = 0$, $V = 0$, $W = 0$ are particular line quadric surfaces through the six points then the equation of the general quartic surface through the six points will be of the form

$$aU^2 + 2bUV + cV^2 + 2dW + e = 0;$$

where $(a,\ b,\ c,\ d,\ e)$ are constants. In order to that there is a node we must have for the coordinates of the vertex, the equations combined with the given equation

$$\begin{aligned}
a\frac{dU}{dx} + b\frac{dV}{dx} + b\frac{dW}{dx} + d\frac{dT}{dx} &= 0,\\
a\frac{dU}{dy} + b\frac{dV}{dy} + c\frac{dW}{dy} + d\frac{dT}{dy} &= 0,\\
a\frac{dU}{dz} + b\frac{dV}{dz} + c\frac{dW}{dz} + d\frac{dT}{dz} &= 0,\\
a\frac{dU}{dw} + b\frac{dV}{dw} + c\frac{dW}{dw} + d\frac{dT}{dw} &= 0.
\end{aligned}$$

Eliminating $(a,\ b,\ c,\ d)$, we have an equation $\nabla = 0$, where ∇ is the Jacobian or fractional determinant $\dfrac{d(U,\ V,\ W,\ T)}{d(x,\ y,\ z,\ w)}$ formed with the differential coefficients of the four functions $(U,\ V,\ W,\ Z)$; the locus of the vertex is then a quartic surface.

Calling the six points 1, 2, 3, 4, 5, 6, then taking as vertex any point in the line 12, the lines from such point to the points 1 and 2 coincide with the line 12, and we can through this line and the lines to the remaining points 3, 4, 5, 6 describe a quadric cone; the quartic surface therefore passes through the line 12; and similarly it passes through each of the fifteen lines 12, 13, …, 56.

Again, taking the vertex anywhere in the line of intersection of the planes 123 and 456, we have an improper quadric cone, viz. the plane-pair formed by these two

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planes; the line in question is therefore a line of the quartic surface, and similarly the quartic surface contains all the two lines 123, 456; 124, 356; 134, 456. We have thus in all 25 lines on the quartic.

In the case of seven points 1, 2, 3, 4, 5, 6, 7, the locus is the curve of inter- section of the quartic surfaces (obtained through any one of the six points 1, 2, 3, 4, 5, 6; 23, 24, 25, 26, 34, 45 to which it should be $= 16$, $b = 71$, $c = 56$). Here the several lines on either of the surfaces, viz. part of the curve in question are: viz. on or scarce curve, which is the locus of the vertices of the cones which pass through the seven given points.

14. *Show that the envelope of a system of circles having its centre on a given and cutting at right angles a given circle is a bicircular quartic; which, when the plane which and circles has double contact, becomes a pair of circles; and, by considering the magnitude of the latter, deduce, the area of curve in regard between the two principal planes.*

(1) given two circles, to find a polygon of n sides inscribed in the one and circum- scribed about the other.

The equation of the given circle is taken to be

$$(x-a)^2 + (y-\beta)^2 = a^2,$$

and that of the conic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. This being so, we have $a\cos\theta$, $b\sin\theta$ as the coordinates of a point on the conic, which point may be taken to be the centre of the variable circle, and introducing the condition that the two circles cut at right angles, the equation of the variable circle is

$$(x - a\cos\theta)^2 + (y - b\sin\theta)^2 = a^2\cos^2\theta + b^2\sin^2\theta + c^2,$$

that is,

$$x^2 + y^2 - a^2 - 2xa\cos\theta - 2yb\sin\theta + b^2\sin^2\theta - 2yb\sin\theta = 0,$$

where θ is the variable parameter; and the equation of the envelope therefore is

$$(x^2 + y^2 - c^2)\cos^2\theta + (a^2 - b^2)\sin\theta\cos\theta - 2ax\cos\theta - 2by\sin\theta + b^2 = 0,$$

which is a quartic, having and belonging to each as b, c, in the same equation as would have been obtained from either of the other conics and the corresponding circle, and from the form of the equation it is clear that the curve is a bicircular quartic.

If the fixed circle touches the conic, then by a consideration of the figure it at once appears that the point of contact is a double point on the curve; and as if there is a double contact, then each of the points of contact is a double point on the curve. But in this case the curve is a bicircular quartic with four double points, viz. it is a pair of circles.

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The points in regard to the two conics is that, in general it is not possible to find any polygon of n sides satisfying the conditions; but that the case is also such that there exists an infinity of polygons; viz. any point whatever of the one conic may then be taken as a vertex of the polygon, and there are corresponding the the $(n+1)^{\text{th}}$ vertex still coincide with the first vertex, and there will be a polygon of n sides.

Now imagine that the conic touched by the sides is a circle having double contact with the other conic. Describe any one of the polygons, and with each vertex as centre describe the orthotomic circle, which will, it is clear, be a circle passing through the points of contact with the circles, each touching the two adjacent circles of the series. And by considering any other polygon, we have a like series of a circles; and by what precedes the envelope of all the circles of the several series is a pair of circles; that is, the circles of every series touch these two circles. We have consequently two circles, such that there exists an infinity of closed series of n circles, each circle touching the two fixed circles, and also the two adjacent circles of the series; which is the porism arising out of the second problem.

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Note on the Problem of Envelopes.

[From the Messenger of Mathematics, vol. I. (1872), pp. 3, 4.]

THERE is a mode of looking at the problem of Envelopes, which, so far as I am aware, has not been explicitly noticed. Let $U = (x, y, z)^m$ be a function of the coordinates (x, y, z) $(\Theta = 0$ or (x, y, z) of θ, a function of the new set of coordi- nates (x, y, z) and (x', y', z'), θ being understood that when we write $\Theta = 0$ this expresses the existence of a relation between the coordinates (x, y, z) and the new set of coordi- nates (x', y', z'). Suppose that we have $\Theta = 0$, $U = 0$; that is, then the equation expresses θ as a function of (x, y, z), (x', y', z'); this is then a curve: the equation expressed values of parameters in coordinates (x, y, z) of a point P on the curve $U = 0$; and we may say that θ is the envelope (equivalent to four equations only)

$$U = 0, \quad \Theta = 0,$$

$$d_x U + \lambda d_x \Theta = 0, \quad d_y U + \lambda d_y \Theta = 0, \quad d_z U + \lambda d_z \Theta = 0,$$

and a particular integral $ay - b = 0$: the complete integral is in fact

$$(x - cy)^2 - 4r^2b^2 + 2(p - q)y^2 - rp^2 = 0,$$

satisfied, for $C = 0$, by $xy - n = 0$.

But, observe that the required envelope is the locus of the points of intersection of the curve $U = 0$, by curves in a particular point (x', y', z') of the other curve, $\Theta = 0$. So the curve $U = 0$ which belongs to a consecutive point (x', y', z') of the curve $U = 0$, con- sidering therein (x', y', z') as the variable; by such a process the envelope of the curve y in question is a mere definition of what it does mean, and comes to be a fluxion. y then only chosen the values of A are b, b', b'', &c., and these values are then determined by the equations:— viz. if P is a point on the conic (x', y', z'), then conics, conditions (x', y', z') are conics on the same conic (x', y', z') as their values: and the curve in the plane which corresponds spend to the generating loop of the model.

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$\Theta = 0$, $U = 0$ to touch each other, we have a relation in (x', y', z') which is the equation of the point intersection of the curves $\Theta = 0$ belonging to the two consecutive values of the curve $U = 0$; that is, the equation of the required envelope is obtained as the condition that the curves $U = 0$, $\Theta = 0$ shall touch each other. But when the curves touch each, they have at the point of contact two consecutive points proportional, or we have simultaneously

$$U = 0, \quad \Theta = 0,$$

$$d_x U + \lambda d_x \Theta = 0,$$
$$d_y U + \lambda d_y \Theta = 0,$$
$$d_z U + \lambda d_z \Theta = 0,$$

the same equations as before, since $\Theta = 0$ and $\Theta = 0$ are the same function.

It is to be added that, when $a = u$, the equations

$$d_x U + \lambda d_x \Theta = 0,$$

$$d_y U + \lambda d_y \Theta = 0,$$
$$d_z U + \lambda d_z \Theta = 0,$$

are homogeneous in (x, y, z), and we may by the elimination of (x, y, z) from these equations obtain an equation Disc. $(U \pm \lambda U) = 0$, say for shortness $\Delta = 0$, involving λ and also the coordinates (x', y', z'). Now it is a known theorem that the condition for the contact of the two curves $U = 0$, $U = 0$ may be obtained by expressing that the equation $\Delta = 0$ shall have a pair of equal roots, or, what is the same thing, by equating to the equation of the envelope of the curve $U = 0$, viz. by being so as shown the equation of the envelope of the curves $U = 0$, is in fact

Disc. $\text{Disc.}_{(\lambda,\,1)}(U + \lambda U) = 0$;

viz. we first take the discriminant of the function $\Theta + \lambda U$ in regard to the coordinates (x, y, z) and then taking the discriminant in regard to λ of the function so obtained, b is zero. This is in many cases a more simple process than that of the direct elimination of x, y, z, λ from the equations.

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Note on Lagrange's Demonstration of Taylor's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), pp. 22–24.]

I TAKE the occasion of the publication of the last edition of Mr Todhunter's *Treatise on the Differential Calculus* to make some remarks on the demonstration in question. Mr Todhunter proposes to himself to establish a comprehensive view of the *Differential Calculus* as the method of Limits; but he very properly introduces in some cases demonstration founded upon other views of the subject, pointing out that this is the case, and explaining or justifying his objections. Thus (Chapter VI.) upon Taylor's Theorem, he remarks: "Before we offer a strict demonstration of the theorem in question, we shall notice the method which it was usual to adopt in treatises on the Differential Calculus not based on the doctrine of limits;" and then, after giving a demonstration depending on the relation $\dfrac{d}{dx}f(x+h) = \dfrac{d}{dh}f(x+h)$ he goes on to say "There are numerous objections to the method of the preceding article, and especially the use of an infinite series, without ascertaining that it is convergent, is inadmissible; we proceed then to prove Taylor's Theorem after the manner adopted by Mr Homersham Cox(*) in a demonstration of the equation

$$f(x+h) = f(x) + hf'(x) + \cdots + \frac{h^n}{[n+1]^{n+1}} f^{(n+1)}(x+\theta h),$$

(θ between 0 and 1) Taking "if the function $f^{(n)}(x+\theta h)$ is such that by making a sufficiently great the term $\frac{h^n}{[n+1]^{n+1}} f^{(n+1)}(x+\theta h)$ can be made as small as we please, then by carrying on the series

$$f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \&c.$$

* This demonstration is similar in principle to Laprange's but I think it is preferable; do, the gives in passing in this rather than the same value whether n is changed once, ; it on it. VIII, p. 9 (1851).

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to as many terms as we please we obtain a result differing as little as we please from $f(x+h)$. Under these circumstances then we may assert the truth of Taylor's theorem."

I share Abel's horror of divergent series(*); and I maintain the validity of Lagrange's demonstration. When by any objection one can expand a function in a series, for instance the function $\frac{1}{1-x}$, by division

$$\frac{1}{1-x} = 1 + x + \frac{x^2}{1-x} = 1 + x + x^2 + \frac{x^3}{1-x}, \&c.$$

$$1 + x + x^2 + x^3 + \&c.,$$

in the series $1 + x + x^2 + \&c.$, and write accordingly

$$\frac{1}{1-x} = 1 + x + x^2 + \&c.$$

all that is for ought be may mean is that the algebraical operations contained as far as we please will give the series of terms 1, x, x^2, &c., or say the series of coefficients 1, 1, 1, ... And of course with this meaning of the equation, the objection "we cannot that the series is convergent" would be wholly irrelevant, we do not say that in is, we do not care whether it is &c. or not. In further illustration, remark that we frequently use such an equation merely as the means of expressing the law of a series of numbers a_0, a_1, a_2, &c., $a_n = a + b\theta$, &c.; we make the assertion to be a definite process expressible in the form $a_n =$ co. x^n in $a_n = a_0 + a_1 x + a_2 x^2 + \&c.$ in question. Any objection that the series is not convergent would be simply irrelevant. Now an that the series shall be convergent is wholly employed for the expansion of $f(x+h)$ in regard to $f(x+h)$. It is impossible, in a quantitative algebra such as is presupposed in the method of limits, to put any meaning on the formula

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \&c.,$$

viz. $f(x+h)$ requires the same value $f(x+h)$ whether we change therein a into $x+h$ or h into $x+h$; and the expression on the right-hand side is the only series in h possessed of the same property. It is to be remarked that the equation contains in itself the definition of the operation of derivation, viz. the equation being true, $f(x)$ can only denote the coefficient of h in the expansion of $f(x+h)$, and what

* Note on Aragon dans des bons louisges dipres divines.

$$------ - \theta + - (-\theta)^2 + \&c. ------$$

a maint et avait ferois priselt ?—r–, t. IV. p. 263; [Newt. Eli, 1885, t. VII. p. 122].

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really is shown is that admitting such an operation to be possible in regard not only to $f(x)$, but to $f'(x)$, &c., then the coefficients $f'(x)$, $\frac{f''(x)}{2!}$, &c., are obtained from $f(x)$ by the successive repetition of this operation and by dividing by the proper numerical denominator.

By what precedes, any objection in regard to convergency, I regard as irrelevant; and if it is said that the above-mentioned single assumption is not granted, I would either ask "What is a Function?"—or I would contend myself with the hypothetical statement: "If $f(x)$ be such that $f(x+h)$ is expansible in apoly, then Taylor's theorem.

In regard to the demonstration given by Mr Todhunter, it impliedly assumes that x and h are both real, and (although doubtless possible) it would be considerably more difficult to find an analogous demonstration of the formula involving $f^{(n+1)}(x+\theta h)$ in the case of x and h imaginary. But the formula with the term in question is not (see Mr Todhunter consider it as being) Taylor's theorem, ; to obtain from it Taylor's theorem, we require (in the foregoing point of view) the property that $\lim_{n\to\infty} f^{(n+1)}(x+\theta h)$ is exposable in a series involving h^{n+1} and the higher powers of h, that is, the very property that $f(x+h)$ is exposable in positive powers of h.

Moreover admitting that the formula with the term $f^{(n+1)}(x+\theta h)$ is demonstratable for imaginary values of x, h, then formula is unenchapsie in the case where $x+h$ is one or both a symbol or symbols of operation; whereas it would certainly have to include unmoral magnitude, and if it is considered as meaning anything, then the equation in question is mere definition of what it does mean, and ceases to be a theorem in regard to $f(x+h)$. It is impossible, in a quantitative algebra such as is presupposed in the method of limits, to put any meaning on the formula

$$f\left(\frac{d}{dx}+h\right) = f\left(\frac{d}{dx}\right) + hf'\left(\frac{d}{dx}\right) + \&c.,$$

which however I regard as a legitimate particular form of Taylor's theorem.

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Solutions of a Smith's Prize Paper for 1871.

[From the Messenger of Mathematics, vol. I. (1872), pp. 37–47, 71–77, 89–95.]

1. *A point moves in a plane with a given velocity, and also with a given angular velocity about a fixed point in the plane: show that the locus is either a circle passing through the fixed point, or else a circle having the fixed point for its centre; and explain the relation between the two solutions.*

We have in general

$$v^2 = \left(\frac{dr}{dt}\right)^2 + r^2\left(\frac{d\phi}{dt}\right)^2,$$

and in the present question, taking the fixed point as the origin, and measuring θ from any fixed line through this point,

$$\frac{d\theta}{dt} = a, \quad V^2 = \left(\frac{dr}{dt}\right)^2 + r^2a^2,$$

where V, a are given constants. Hence

$$\left(\frac{dr}{dt}\right)^2 + a^2r^2 = V^2, \quad \left(\frac{dr}{dt}\right)^2 = V^2 - a^2r^2,$$

or, writing $V = aa$,

$$\left(\frac{dr}{dt}\right)^2 = a^2(a^2 - r^2);$$

therefore

$$d\theta = \frac{dr}{\sqrt{(a^2 - r^2)}},$$

or

$$\theta + \beta = \sin^{-1}\frac{r}{a}, \quad (\beta \text{ the constant of integration,})$$

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that is,

$$r = a\sin(\theta + \beta),$$

which is the equation of a circle (radius $= \frac{1}{2}a$) passing through the fixed point. In fact, the point moving in such a circle with a constant velocity, moves about the centre with a constant angular velocity; and about any fixed point in the circumference with an angular velocity which is one-half that about the centre, and is therefore also constant.

Treating θ as a variable parameter, to obtain the envelope we have

$$0 = a\cos(\theta + \beta),$$

that is, $\theta + \beta = \frac{\pi}{2}$, and therefore $r = a$, which is the equation of a circle (radius $= a$) having the fixed point for its centre. This is consequently the singular solution.

2. *Determine the system of curves which satisfy the differential equation*

$$dx(\sqrt{(1 + x^2 + xy + xy^2)}) + dy(\sqrt{(1 + x^2 + xy + xy^2)}) = 0;$$

and that the curve which passes through the point $x = 0, y = 0$, *contains as a part of itself the conic*

$$x^2 + y^2 + 2xy\sqrt{(1 + x^2 + xy + xy^2)} = 1.$$

The equation is integrable *per se*, viz. we have

$$x\sqrt{(1 + x^2)} + \log\{x + \sqrt{(1 + x^2)}\} + \frac{1}{2}y\sqrt{(1 + y^2)} + \log\{y + \sqrt{(1 + y^2)}\} = U$$

or, denoting the constant so that the differential equation has its general solution

$$x\sqrt{(1 + x^2)} + y\sqrt{(1 + y^2)} + 2xy = (1 + x^2 + xy + y^2) + \log\frac{\{x + \sqrt{(1 + x^2)}\}\{y + \sqrt{(1 + y^2)}\}}{1 - C} = 0,$$

which is evidently a transcendental curve; it may however be shown that, if

$$x^2 + y^2 + 2xy\sqrt{(1 + x^2 + xy + xy^2)} - 1 = 0,$$

so that the equation of the curve is then made obvious, that is, the transcendental curve $y = axy + b$, which is the elementary curve, is in this manner included as a part of the curve.

[At a simple illustration as to how this may happen, take the transcendental curve $y = axy + b$, which is the elementary curve, in this manner included as itself.]

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We have, from the equation of the conic

$$\{x+y\sqrt{(1+x^2)}\}^2 = (1+x^2)(1-y^2),$$

that is,

$$x+y\sqrt{(1+x^2)} = \pm\sqrt{(1+x^2)(1-y^2)},$$

but considering the radicals as positive, the sign must be taken so that we have simultaneously $x+y>0$, $y=a$. We have therefore

$$x+y\sqrt{(1+x^2)} = \sqrt{(1+x^2)(1-y^2)},$$

and similarly

$$y+x\sqrt{(1+y^2)} = \sqrt{(1+y^2)(1-x^2)}.$$

Then

$$x\{x+y\sqrt{(1+x^2)}\}+y\sqrt{(1+x^2)} = 2xy+(x^2+y^2)\sqrt{(1+x^2)}+x\sqrt{(1+x^2)(1-y^2)}$$

$$= 2xy+(x^2+y^2)\sqrt{(1+x^2)}-2xy(1+x^2)$$

$$= xy(1-x^2-y^2)+\sqrt{(1+x^2)}(x^2+y^2)-2xy(1+x^2)$$

$$= xy(1-x^2-y^2)-2xy(1+x^2),$$

which is the first of the relations in question; and similarly

$$x\{y\sqrt{(1+x^2)}+x\sqrt{(1+y^2)}\} = \sqrt{(1+y^2)}+y\sqrt{(1+x^2)}$$

$$= y(1+y^2)-2y(1-x^2-y^2)$$

$$= (1+y^2)\sqrt{(1+x^2)}-2y(1-x^2-y^2)\sqrt{(1+x^2)}$$

$$= -(x+y\sqrt{(1+x^2)})(x+y\sqrt{(1+y^2)}),$$

which is the second of the two relations. And the theorem is thus proved.

The foregoing is the easiest and most obvious solution, but it is interesting to consider the question differently, as follows:

Write

$$Q = \frac{(x+y\sqrt{(1+x^2)})(y+x\sqrt{(1+y^2)})}{x+y\sqrt{(1+x^2)}};$$

we have

$$Q = (x^2+y^2+xy\sqrt{(1+x^2)}+xy\sqrt{(1+y^2)}+y\sqrt{(1+x^2)(1+y^2)})-A-B,$$

$$Q^* = (x^2+y^2+4)(\sqrt{(1+x^2)}+\sqrt{(1+y^2)}+x\sqrt{(1+x^2)(1+y^2)}) = A-B,$$

if

$$A = x(1+x^2)(1+y^2)+2xy,$$

$$B = x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}+xy\sqrt{(1+x^2)(1+y^2)}),$$

and then

$$AB = xy\sqrt{(1+x^2)}+xy\sqrt{(1+y^2)}+y\sqrt{(1+x^2)(1+y^2)},$$

$$+xy(1+y^2)\sqrt{(1+x^2)}+(1+x^2)(1+y^2)\sqrt{(1+x^2)}+xy\sqrt{(1+x^2)(1+y^2)},$$

$$= xy\sqrt{(1+x^2)}+xy\sqrt{(1+y^2)}+2xy\sqrt{(1+x^2)(1+y^2)}.$$

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that is,

$$Q^2\{2xy+1+2x\sqrt{(1+x^2)}\} - \tfrac{1}{2}xy\{2xy+1+2x\sqrt{(1+x^2)}\} = 0,$$
$$= 4(x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)})+4xy$$
$$+4xy\{Q\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}-\tfrac{1}{2}\sqrt{(1+x^2)(1+y^2)}\};$$

whence

$$4\{x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}+2xy\} = Q^2(2xy+1+2x\sqrt{(1+x^2)})$$
$$= Q^2\{2x\sqrt{(1+x^2)}+1+2x\sqrt{(1+x^2)}\} - \tfrac{1}{2}xy\{2x\sqrt{(1+x^2)}+1+2x\sqrt{(1+x^2)}\}$$
$$+4xy(Q\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}-\tfrac{1}{2}\sqrt{(1+x^2)(1+y^2)})$$
$$= (Q-1)(2xy+1+2x\sqrt{(1+x^2)})$$
$$+4(Q-1)xy\sqrt{(1+x^2)}$$
$$-4(Q-1)(1+xy^2)$$
$$-4(Q-1)xy^2,$$

that is,

$$x\sqrt{(1+x^2)}+y\sqrt{(1+y^2)}+2xy = (Q-1)D.$$

And the integral equation is

$$\tfrac{1}{2}(Q-1)+\log Q = C,$$

which, for $C=0$, is satisfied by $Q-1=0$.

Now starting from

$$Q = \frac{(x+y\sqrt{(1+x^2)})(y+x\sqrt{(1+y^2)})}{x+y\sqrt{(1+x^2)}}$$

we have

$$\sqrt{(1+x^2)}-x = \tfrac{1}{2}y\{\sqrt{(1+x^2)}-x\}+\tfrac{1}{2}\{\sqrt{(1+y^2)}-y\},$$

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and thence

$$2x = K\sqrt{(1+x^2)} + Ly,$$

if

$$K = Q\sqrt{(1+x^2)} + x + \tfrac{1}{2}\sqrt{(1+x^2)-1},$$
$$L = Q\sqrt{(1+y^2)} + y + \tfrac{1}{2}\sqrt{(1+y^2)-1},$$

whereon

$$L^2 - K^2 = 4.$$

Moreover

$$(2x+Ly)L = K^2 + KLy$$

that is,

$$4x^2 + Ly = K^2 + KLy = K^4,$$

or, what is the same thing,

$$x^2 + Lxy = (K-x)^2,$$

which is the rationalized form

$$Q = \frac{x^2 + 2xy\sqrt{(1+x^2+xy+y^2)} - y^2}{xy + \sqrt{(1+x^2)(1+y^2)}} - 1.$$

And if $Q = 1$ then $L = 2\sqrt{(1+x^2)}$, $(L - K^2 = 2x)$, and the equation is

$$x^2 + 2xy\sqrt{(1+x^2)} - y^2 = 0,$$

that is, the curve $C = 0$, of the integral integral has, as is should be, a part the conic

$$\frac{x^2 - y^2}{x + y\sqrt{(1+x^2)}} = 1, \quad \text{or as it should be a fixed conic.}$$

We may evidently differently variations, and present the results as follows: the equation

$$\frac{1}{2}\left(\frac{x}{a}+1\right)\left(\frac{x}{a}-1\right) = \frac{Q+1}{2}\left[\frac{x}{a} + \frac{y-K}{a}\right] - C$$

and a particular integral $xy - n = 0$: the complete integral is in fact

$$(x-cy)^2 - 4r^2e^2 + 2(p-q)y^2 - rp^2 = 0$$

satisfied for $C = 0$, by $xy - n = 0$.

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3. Write $x = b - c$, $\beta = c - a$, $\gamma = a - b$; then considering the three circles and the three conics

$$(x-a)^2 + y^2 = 2bc, \quad \frac{x^2}{Y^2} + \frac{Y^2}{K+bc} = 1,$$

$$(x-b)^2 + y^2 = -2ac, \quad \frac{x^2}{K+bc} + \frac{Y^2}{K+ca} = 1,$$

$$(x-c)^2 + y^2 = 2ab, \quad \frac{x^2}{K+bc} + \frac{Y^2}{K+ab} = 1.$$

where K is arbitrary; it is required to show that of a variable circle having its centre in one of the conics cuts at right angles the corresponding circle, the envelope is in each of the three cases one and the same bicircular quartic.

Consider the circle $(x-a)^2 + y^2 = 2bc$ and the conic $\dfrac{X^2}{Y^2} + \dfrac{Y^2}{K+bc} = 1$, the coordinates of a point on the conic are $\cos\theta\sqrt{(K)}$, $\sin\theta\sqrt{(K+bc)}$, where θ is a variable parameter; say for a moment these values are p and q. Hence, taking (p, q) as the centre of the variable circle the equation is

$$(x-p)^2 + (y-q)^2 = (p-a)^2 + q^2 - 2bc,$$

and in order that this may cut at right angles the circle

$$(x-a)^2 + y^2 = 2bc,$$

we must have

$$(p-a)^2 + q^2 = 2(p-a)^2 - 2bc + 2bc,$$

or, substituting for p and q their values, the equation of the variable circle is

$$(x-p)^2 + (y-q)^2 = pq^2 + q^2 - 2bp,$$

that is,

$$x^2 + y^2 - p^2 - 2p\sqrt{(K)} - 2q\sqrt{(K+bc)} + 2yp = 0,$$

viz. this is

$$(x^2 + y^2 - p^2)^2 - 2(x+a)\sqrt{(K)}\cos\theta + 2y\sqrt{(K+bc)}\sin\theta + p^2 = 0.$$

Hence taking the envelope in regard to θ, the equation is

$$(x^2 + y^2 - p^2)^2 = 4K(x+a)^2 + 4(K+bc)\,y^2 - 4y\sqrt{(K+bc)}(x+a) + 4Kp^2,$$

or, what is the same thing,

$$[(x+y)^2 - 2a(x+y)\sqrt{(K)} + bc(K+bc)y^2 - 4Ky^2 + 4Kp(2x-bc) = 0,$$

viz. this equation, being symmetrical in regard to a, b, c, is the same equation as would have been obtained from either of the other conics and the corresponding circle; and from the form of the equation it is clear that the curve is a bicircular quartic.

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4. *Show that the caustic by reflexion for parallel rays of a circle, radius r, index of refraction μ, is the same curve as the caustic by refraction for parallel rays of a concentric circle, radius $\frac{r}{\mu}$, index of refraction $\frac{1}{\mu}$.*

Take a centre $\mu > 1$. Imagine the ray AP (fig. 1) parallel to the axis of x, incident at P on the circle radius r, and let the reflected ray after cutting the circle, radius $\frac{r}{\mu}$, cut it again at Q, and then cut the axis in R. Take ϕ, ϕ' for the angles of incidence and refraction, viz. $\angle O = \phi$, that is in the manner suitable for the caustic by reflection.

Fig. 1.

Moreover on the triangle PQO, we have $\sin Q : \sin\phi = r : \frac{r}{\mu}$; that is, $\sin Q = \mu \sin\phi$, $= \mu \sin\phi' = \sin\phi'$; or $\angle Q = \phi'$. And then in the triangle RQO, $\angle R = \phi - \phi'$, or $\angle ORQ = 180° - \phi$.

Consider now a ray BQ incident at Q and refracted in the direction QR; the index of refraction being $\frac{1}{\mu}$, that is, the denser medium being on the outside of small circle. Taking θ, θ' for the angles of incidence and refraction, we have $\theta = \frac{1}{2}\sin^{-1}\phi'$, $= \mu\sin\phi'$ that is, $\theta = \phi$; the angles being two pencils of rays each corresponding to angle BQ of the second pencil, the rays AP and BQ each giving rise to the same refracted ray PQR; hence the two pencils have the same caustic.

[It is proper to remark that for the ray BQ it has been assumed that the refraction takes place not at Q where it first meets the small circle, but at Q.

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we consider the refraction at Q, then the index of refraction is still to be $= \frac{1}{\mu}$, that is, the denser medium must be inside the small circle: the refracted ray is in the direction RQ situate symmetrically with RQ on the opposite side of the axis of y; and it would at first sight appear that this would be a curve equal and similar to the original caustic, but situate on the opposite side of the axis of y. But geometrically the complete caustic consists of two equal and similar arches situate on the opposite sides of the axis of y; so that we really obtain, not an equal portion of the caustic, but the same caustic over again (about the limit changed, viz. as below by (ΔU). Substitute the variation of U in the symbols of operation differs; the formula is, the variations of Z, then the radii in radii of refraction is still to be $= \frac{1}{\mu}$, that the formula deduced from the variation of Z in so far as it arises from the variation of the constants, viz. the equation $\Delta U = \frac{dU}{d\alpha} d\alpha + d\beta = 0$ remains.

There are the like equations in regard to y, z; viz. in all, four equations which are linear in regard to $\frac{d\alpha}{d\alpha}$, $\frac{d\alpha}{d\beta}$, $\frac{d\alpha}{d\gamma}$, $\frac{d\alpha}{d\delta}$, and which serve to determine these quantities in terms of α, β, γ, δ.

Now considering the a, y as expressed in terms of a, c, e, …, then Ω becomes a function of these quantities; the differential coefficients $\frac{d\Omega}{d\alpha}$, &c., being connected with the original differential coefficients $\frac{d\Omega}{d\alpha}$, $\frac{d\Omega}{d\beta}$, &c. by the equations

$$\frac{d\Omega}{d\alpha} = \frac{d\Omega}{d\alpha} \cdot \frac{d\alpha}{d\alpha} + \frac{d\Omega}{d\beta} \cdot \frac{d\beta}{d\alpha}, \quad \&c.$$

As there are four equations, $\frac{d\Omega}{d\alpha}$, $\frac{d\Omega}{d\beta}$, &c. can be expressed in terms of a, c, e in an infinity of ways in terms of $\frac{d\Omega}{d\alpha}$, $\frac{d\Omega}{d\beta}$, &c., and, considering $\frac{d\Omega}{d\alpha}$, &c., as given in terms of $\frac{d\Omega}{d\alpha}$, &c., we can in an infinity of ways express $\frac{d\Omega}{d\alpha}$, &c., as linear functions of $\frac{d\Omega}{d\alpha}$, $\frac{d\Omega}{d\beta}$, &c. But there is no form (obtained by combining the equations in a particular manner) wherein the coefficients of the non-constant quantities are functions of a, c, e, … without a contact; and this is the form most simply employed for the expression of $\frac{d\Omega}{d\alpha}$, &c. in terms of $\frac{d\Omega}{d\alpha}$, $\frac{d\Omega}{d\beta}$, &c. in the method wherein these quantities are made use of in this respect.

I remark upon the present question, that the answer ought to be in evidence perfectly familiar to every student in *Physical Astronomy*; and that a student ought to be able to present it in a clear and lucid form: the question being in fact intended as a test of ability in this respect.

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But in virtue of the relation $a' + \beta' + \gamma' = 1$, we have

$$a\frac{dT}{dx} + \beta\frac{dT}{dy} + \gamma\frac{dT}{dz} = 0,$$

$$a\frac{dy}{dx} + \beta\frac{dy}{dy} + \gamma\frac{dy}{dz} = 0, \quad \&c.$$

Hence subtracting the corresponding equations we have three equations, which are at once seen to be equivalent to the two equations

$$\frac{dT}{dx}\,d\alpha + \frac{dT}{dy}\,d\beta + \frac{dT}{dz}\,d\gamma = 0, \quad \beta : \gamma,$$

equations which must be satisfied identically, whatever are the values of $(\alpha,\, y,\, z)$. The equations have been obtained as necessary conditions; they are, in fact, the sufficient conditions for a double system; for the line being supposed in passing from P to P', to remain always in contact with the surface, the result is in passing from P to P', it is necessary that the same line should also be a line of the surface.

6. If particles describe a side of the form $(a,\, b,\, \ldots \breve{a},\, x,\, 1)^n = 0$ with arbitrary coefficients, and if equation $ax^2 + bXy + cy^2 = 0$, where of $(a,\, b,\, \ldots)$ are arbitrary coefficients respectively in any manner whatever in the function G, then we have a series of equations satisfied by the values x, y which belong to the double root.

The equations

$$aXX + 2bXY + cY^2 + 2(pX + qY) + r = 0$$

$$EX + 2tY + s = 0$$

of common to the two conics, but for this equation to be the case of Z as a W shall vanish, the relation in the points of the plane $\dfrac{X^2}{a} + \dfrac{Y^2}{b} + \dfrac{Z^2}{c} = 1$; and that X, Y, Z, W are quartic functions of $(x,\, y,\, z)$ a common roots, show that this condition of X being equal to V, W, that is, changing the sign in the same way as the linear function of $(x,\, y,\, z)$ rationally, and consequently each of the original four roots, may be represented by an equation of the form.

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Taking the conics to pass through the point $a = 0$, $y = 0$; their equations will be of the form

$$X = ax^2 + 2hxy + by^2 + 2fx + 2gy + qy = 0,$$

$$Y = ax^2 + 2hxy + by^2 + 2fx + 2gy + qy = 0,$$

$$Z = ax^2 + bxy + cy^2 + 2dy + ey^2 + qy = 0,$$

$$W = ax^2 + bxy + cy^2 + 2dy + ey^2 + qy = 0.$$

Now multiplying by the indeterminate quantities α, β, γ, δ; their equations will be of the form. We can determine the values of a, β, γ, δ so that in the sum $aX + \beta Y + \gamma Z + \delta W$ the coefficients of x, y, 1 shall vanish, and the term in xy shall, that is, in $\alpha\beta\gamma\delta$, of values of a, β, γ, δ, but as the case may a quartic equation for the values of the ratios; and say $\alpha\beta\gamma\delta : \beta\gamma\delta : \gamma\delta : \delta$ changing the coordinates (x, y), this remaining equation may be represented by $a^2 = 0$.

And it is clear that we then have, in the system of conics $aX + \beta Y + \gamma Z + \delta W = 0$, four conics the equations of which may be assumed to be of the form

$$X = ax^2 + ax,$$

$$Y = ax^2 + ax,$$

$$Z = ax^2 + ax,$$

$$W = a^2x + ax^2,$$

so that, by writing $a = ax + \cdots$, and f, f', f'', f''' are the same conic upon the same linear function of (x, y, z) rationally, and consequently each of the original four roots, may be represented by an equation of the form.

7. *The coordinates (x, y, z) of a point P are given as connected with the coordinates (x', y', z') of a point P' on a plane by equations*

$$x : y : z = a' : Y' : Z' : W';$$

whereof showing, we have

$$ax^2 + y' = 0 :$$

Discuss the connection of P' and P in regard to the equations of points and curves on the two surfaces.

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The equations $x : y : z : w = X' : Y' : Z' : W'$ are two equations establishing indeterminate parameters $x' : y' : z'$, &c. that the eliminating these we have between $(x,\, y,\, z)$ a single (homogeneous) equation representing a surface. To each point $(x,\, y,\, z)$ a single value $x' : y' : z'$ corresponds and similarly to each point $(x',\, y',\, z')$ a single (homogeneous) equation representing a surface. To each point $x' : y' : z' : 1$ which is a corresponding point $(x,\, y,\, z)$ on the surface.

To find the order of the surface, consider its intersection with any arbitrary line: ax + by + cz + dw = 0,

$$a'x + b'y + c'z + d'w = 0.$$

We have corresponding equations in the plane of intersection of the surface;

$$aX' + bY' + cZ' + dW' = 0,$$

$$a'X' + b'Y' + c'Z' + d'W' = 0;$$

viz. these are two curves: each of them passing through the common point of the four curves, and therefore having common besides these points, three other points: the order of the surface is therefore = 3.

To show that the common point ought to be the (as above) excluded, some further consideration is necessary. To the points of the surface $aX + bY + cZ + dW = 0$ corresponds the locus $aX'+bY'+cZ'+dW' = 0$ and similarly to the points of the linear curve $aX'+bY'+cZ'+dW' = 0$ corresponds the points of the surface $aX + bY + cZ + dW = 0$, then the corresponding loci on the other surface are conics: hence to a similar to the corresponding points of the surface, and which are not common ones (or any belonging to one or more of the conics) the points $(x,\, y,\, z)$, z' which is of the order $(\alpha,\, y)$ ought to be rejected; the equation in question is thus reduced to $z' = (xz - bX + zZ + xW)\, xs' + \cdots = 0$.

The same result may be obtained and may be further shown that the surface is a a, b, c are the same, the form of the original coordinates $(x,\, y,\, z)$ in $(x,\, y,\, z)$ to the corresponding points (in the form of the plane $bXy + aX' + aY' + bZW + gZW' = 0$).

In a particular case, the more general assertion of expression of $x' : y' : z'$, in terms of $(x,\, y,\, z)$ is to be considered, but every linear function of $(x,\, y,\, z)$ rationally, and consequently any sole conic of $(x,\, y,\, z)$ rationally; in particular if $x'^2 + y'^2 + z'^2 = 0$, and these equations may be such, are connected by a relation which, in regard to a fixed line, the plane through this line of the plane through the points $(x,\, y,\, z)$ and $(x,\, y,\, z)$.

The same may not be obtained, instead of the two equations, that is a single equation $a^2z = 1$, having a double root. The equations are than that the $a^2+b^2+c^2+d^2+e^2+6f^2+12g+4h = 0$.

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are, it is clear, $a = bc - rs$, &c. where a is a variable parameter; and accord- ingly the place are than as $r =$, $s =$, suppose. With arbitrary parameter; and through the common intersection of the curve $\Omega = 0$.

If f, V, W are binary functions of the form $(\alpha,\, b,\, \ldots \breve{a},\, x,\, 1)^n$ with arbitrary coefficients, and if the equation $U = 0$, $V = 0$ have a common root, show how this can be determined in terms of the derived functions of the Resultant in regard to the coefficients of either function.

Show what results in regard to the common root can be obtained when the coefficients are not all of them arbitrary but (1) *each or either of the functions depends in any manner whatever on a set of arbitrary coefficients not entering into the other function,* (2) *the two functions depend in any manner whatever on one and the same set of arbitrary coefficients.*

Suppose

$$U = (a,\, b,\, \ldots \breve{a},\, x,\, 1)^m, \quad \left(= ax^m + \frac{m}{1} bx^{m-1} + \&c.\right),$$

$$V = (a,\, b,\, \ldots \breve{a},\, x,\, 1)^n, \quad \left(= ax^n + \frac{n}{1} bx^{n-1} + \&c.\right).$$

Then if R is the resultant, the equation $R = 0$ is the relation which must exist between the coefficients in order that the two functions U, V may have a common root. Moreover, if we assume that U and V may have a common root $x = \alpha$, then we have $U_\alpha = 0$, $V_\alpha = 0$; the coefficients of U_α, V_α being the same as those of U, V, and $x = \alpha$. We have then identically $U_\alpha = 0$, $V_\alpha = 0$, $R = 0$; that is, these equations have to be infinitesimally varied in such manner that U, V have still a common root $x = \alpha$; we have the equation $U = 0$, $V = 0$ subsist when a, b, …, the coefficients of U, and the corresponding parameters of V subsist a corresponding equation in the determinations of the roots a, b, c, &c. or some of them.

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There a series of equations giving the value of the common root $\dfrac{x}{y}(=\alpha)$ in the several ratios

$$\frac{1}{m} : \frac{dR}{da} : \frac{1}{m-1} : \frac{dR}{db} : \frac{1}{m-2} : \frac{dR}{dc} : \frac{1}{1} : \frac{dR}{dk} : \&c.$$

And it is clear that we have in like manner

$$\frac{1}{n} : \frac{dR}{da'} : \frac{1}{n-1} : \frac{dR}{db'} : \&c.$$

It is clear that if U involves, in any manner whatever, the coefficients a, b, ..., which do not enter into the function V, then we have precisely the same expressions as above; and similarly if V involves coefficients which precisely the common root.

A system of equations satisfied by the values x, y which belong to the common root.

But if the coefficients a, b, ... and a', b', &c. are contained in any manner whatever in the function U and V respectively, then we are unable to obtain the common root, either as a rational function of the variables, or by aid of these equations.

11. *Find the differential equation* $\dfrac{dV}{dt} = 0$ *for the absolute force at, a being the distance of the moving system, on the axes of the moving system.*

Taking as coordinates r the radius vector, ω the longitude, y the latitude, the equation of the vis viva function is

$$T = \tfrac{1}{2}(r'^2 + r^2\cos^2 y\,\omega'^2 + r^2y'^2),$$

hence

$$\frac{dT}{dr'} = r' = a \quad \text{suppose},$$

$$\frac{dT}{d\omega'} = r^2\cos^2 y\,\omega' = \nu \quad ,$$

$$\frac{dT}{dy'} = r^2y' = \mu \quad ,$$

and the expression of T in terms of r, a, y and the new coordinates a, ν, μ is

$$T = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right);$$

whence writing

$$H = \frac{1}{2}\left(a^2 + \frac{\mu^2}{r^2} + \frac{\nu^2}{r^2\cos^2 y}\right) + V,$$

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9. *The normal at each point of a principal section of an ellipsoid is intersected by the normal at a consecutive point on the principal section: show that the locus of the point of intersection is an ellipse having four (real or imaginary) common points with the evolute of the principal section.*

The principal section is the convenience taken to be that in the plane of xz, the meridian of the ellipsoid; taking x, z as the coordinates of a point in the plane of xz, and considering the equations of the ellipse; taking a, y, z as the coordinates of a point in the meridian of the ellipsoid; we have for the normal at the point (a, z)

$$\frac{x - x'}{\frac{x'}{a^2}} = \frac{y - y'}{\frac{y'}{b^2}} = \frac{z - z'}{\frac{z'}{c^2}} = -B,$$

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or, substituting for α, the foregoing values,

$$x = x'\left(1 - \frac{B}{a^2}\right), \quad z = z'\left(1 - \frac{B}{c^2}\right);$$

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The principal section in its convenience taken to be that in the plane of xz, the meridian of the ellipsoid; taking x, z as the coordinates of a point in the plane of xz.

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we have

$$\frac{a^2x^2}{c^2} + \frac{a^2z^2}{c^2} = 1,$$

the required form, which is the equation in question.

If the point (X, Y, Z) is on the evolute, it is clear that the corresponding point of the locus will be a point of the evolute of the principal section, and to prove that the locus touches the evolute, it is only necessary to show that the tangent of the locus is also the tangent of the evolute, and which is the same way thing, that the tangent of the locus passes through the umbilics.

Now in the evolute we have

$$X^2 = a^2z'^2 = \rho^2\frac{x'^2}{a^2}, \quad Y^2 = b^2 = \rho^2\frac{y'^2}{b^2};$$

the corresponding values of a, z being

$$x = \frac{a^2X}{p^2}, \quad z = \frac{c^2Z}{p^2}.$$

Take ξ, ζ the current coordinates of a point in the tangent of the locus at a, z; or, substituting for a, z the foregoing values,

$$\frac{X\xi}{a} + \frac{Z\zeta}{b} = 1,$$

and there should be satisfied by $\xi = X$, $\zeta = Z$, or we ought to have

$$\frac{X^2}{a} + \frac{Z^2}{b} = 1,$$

and this equation is in fact true for the values of X, Z in the umbilics; viz. this that is, $\beta - r' = a$, which is in fact the value of β.

There is obviously a contact in each quadrant, that is, there are four contacts the locus passes the umbilics: but if this is the case, then the four contacts will be in the same number of imaginary or rather of i^2.

The same theorem holds good in regard to the principal sections, an ellipsoid by those, the umbilic being imaginary; but the property of contact at the umbilics evolutes are an attraction of the form may be taken to be unitless real, y is then in the probability of a number of heads between a, $a + a$ as one finite quantity; but a, an entirely on number ones in number, the probability of such a number between a and $a + a$ is nearly as may be.

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10. *An evolute being chain of given length is suspended from two fixed points in the same horizontal plane: show that this subject to a condition as to the length of the figure of equilibrium may consist of portions of two distinct catenaries.*

The two parts of the chain will each of them be a portion of a catenary, viz. they will either coincide with each other, forming a twice repeated portion of a catenary which is therefore a possible form of equilibrium, or they will form portions of two distinct catenaries. That the latter form is in some cases possible, appears from the case of a very long chain in which case to the difference of the two positions of equilibrium in which the upper portion is nearly a straight line. In fact, that, in the case of a chain (or rope) just long enough as to reach the ground, there are two forms of equilibrium are equal to a vertex; viz. a vertex, in which the depth, no slope side of the catenary is equal in a regular manner, observe that, in order to the existence of such a form of equilibrium, the necessary condition is, that the tension at A or B must be equal to the two cataries. From this condition, which is the existence of two distinct catenaries, it appears that, when the tension at the joint of a catenary is proportional to the height above the directrix of the catenary, hence the condition is that the directrix of A, and also the line of catenaries through the same directrix, must take the same value the directrix of the directrix.

Take $AB = 2c$, the length of the chain $= 2l$. Take β for the distance of the directrix below the points A, B; c for the parameter of the catenary (or distance of the lowest point above the directrix), β being of course unknown. The taking the origin at the mid-point of the directrix, and the axis of y vertically upwards, the equation of the catenary is

$$y = \tfrac{1}{2}c(e^x + e^{-x}),$$

whence for the point A or B,

$$\beta = \tfrac{1}{2}c(e^c + e^{-c}),$$

and the are measured from the lowest point being

$$s = \tfrac{1}{2}c(e^x - e^{-x}).$$

Hence, assuming that there are two distinct catenaries, if the parameters are c, c', we have

$$\tfrac{1}{2}c(e^x + e^{-x}) + \tfrac{1}{2}c'(e^{x'} + e^{-x'}) = b,$$

$$\tfrac{1}{2}(e^x - e^{-x}) - \tfrac{1}{2}c'(e^{x'} - e^{-x'}) = l,$$

which are the conditions for the determination of a, c', and it is to be shown that these can be satisfied otherwise than by taking $c = c'$.

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Trace the two curves

$$y = \tfrac{1}{2}c(e^{x} + e^{-x}),$$

$$y = \tfrac{1}{2}c(e^{x} - e^{-x}),$$

shown respectively by the black line and the dotted line in fig. 2. Draw any line parallel to the axis of x, meeting the first curve in the points P, P' respectively, and let the ordinates MP, $M'P'$ meet the second curve in the points Q, Q' respectively.

Fig. 2.

then it is clear, that if for a given value of b the line PP' can be drawn in such- wise that $MQ + M'Q' = b$, there will be the required two values $c = OM$, $c' = OM'$, and $c \neq c'$.

And since for MP very large we have MQ diminishes until it attains a finite value, large, and as MP diminishes, $MQ + M'Q'$ also diminishes until it attains a minimum value, PP' has to do then MP' becomes a tangent to the first curve at the vertex; for the two values of the curve PP' being parallel to and below this tangent value, the equation $MQ + M'Q' = b$ is a separate question which is not here considered.

11. *A particle is attracted to two centres of force, one of these on the surface, the other revolving about the origin in a circle in the plane of xy with a uniform angular velocity ν; And the equations of motion, and writing r for the velocity of the particle and A for the constant area (about the fixed centre) in the plane of xy, show that there is a first integral giving the value of $r^2 - 4\nu A\frac{dx}{dy} - g\nu$ in terms of the coordinates of the particle and of the revolving centre.*

Take f, ϕ for the coordinates of the moving centre, so that

$$f = a\cos\nu t, \quad \phi = a\sin\nu t;$$

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the equations of motion are

$$\frac{d^2x}{dt^2} = -P\frac{x}{r} - Q\frac{x-f}{p}, \quad \frac{d^2y}{dt^2} = -P\frac{y}{r} - Q\frac{y-\phi}{p},$$

$$\frac{d^2z}{dt^2} = -P\frac{z}{r} - Q\frac{z}{p},$$

where

$$r^2 = x^2 + y^2 + z^2,$$

$$p^2 = (x-f)^2 + (y-\phi)^2 + z^2.$$

We have

$$r\,dr = x\,dx + y\,dy + z\,dz,$$

$$p\,dp = (x-f)(dx-df) + (y-\phi)(dy-d\phi) + z\,dz.$$

But

$$dt = \nu\sin\nu t\,dt = -d\phi\,dt, \quad df = a\nu\cos\nu t\,dt = d\phi\,dt;$$

whence

$$p\,dp = x\,dx + y\,dy + z\,dz - \omega(\nu dt)dx - \nu(d\phi)dy$$

$$= x\,dx + y\,dy + z\,dz - \nu(p-q)(dx - q\,dy),$$

$$= x\,dx + y\,dy + z\,dz - \nu A(d\phi - q\,dy),$$

that is, the product of the three functions $r^2 - 4\nu A\frac{dx}{dy} - g\nu$, &c. is $= 0$.

Hence from the equations of motion

$$2\left(\frac{dx}{dt}\frac{d^2x}{dt^2} + \frac{dy}{dt}\frac{d^2y}{dt^2} + \frac{dz}{dt}\frac{d^2z}{dt^2}\right)$$

$$= -2P\left(\frac{dr}{r}\right) - 2Q\left(\frac{x\,dx + y\,dy + z\,dz}{p}\right)$$

$$= -2P\frac{dr}{r} - \frac{2Q}{p}\{p\,dp + \nu A(d\phi - q\,dy)\}$$

$$= -2P\frac{dr}{r} - 2Q\frac{dp}{p} + \frac{2\nu A}{p}\{x(d\phi - q\,dy)\};$$

which is in the equation of motion at once $r^2 = -2P\,dr - 2Q\,dp + \frac{2\nu A}{p}\,dx - \frac{2q\nu A}{p}\,dy$

$$v^2 - 2\nu A\left\{x(d\phi - q\,dy) + \frac{2\nu A}{p}(d\phi - q\,dy)\right\} = \int(-2P\,dr - 2Q\,dp).$$

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But we have

$$v^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2,$$

the foregoing equation may be written

$$\nu - 2\nu A\frac{d\phi}{dr} = -2P\frac{dr}{r} - 2Q\frac{dp}{p} + \int \frac{2A}{p}\,dy.$$

whence

$$v^2 - 2\nu A\frac{dq}{dt} = \int \left\{-2P\frac{dr}{r} - 2Q\frac{dp}{p} + \frac{2\nu A}{p}\,dy\right\},$$

the required result.

12. *If a, y are the coordinates of a particle moving in plane under the action of a central force varying as (distance)$^{-3}$; write down the expressions of the constants a_1, a_2 in terms of a and $\dfrac{d^2q}{dt^2}$ of fixed arbitrary constants; and (in case of distorted motion) starting from the equations*

$$\frac{d^2a}{dt^2} = -\frac{P_1}{a^2r^2}, \quad \frac{d^2q}{dt^2} = -\frac{P_2}{q^2r^2},$$

(the notation is be explained), indicate the process of finding the variations of the constants in terms of (1) $\dfrac{d^2a}{dt^2}$, $\dfrac{d^2q}{dt^2}$, (2) *the derived functions of a in regard to the constants.*

We have

$$x = \frac{\cos m - q}{1-\cos m}\,a, \quad y = a\sqrt{(1-r^2)}\frac{\sin m - n}{1-\cos m},$$

where

$$m - n\sin m = \mu(t-t_0) + \nu,$$

an equation serving to express m in terms of the time, and where a, n, t_0, ν; the four- going equations therefore, in effect, give x, y in terms of t and the constants a, r, t_0, n.

In the second part of the question, Ω is a given function of a, y, the differential coefficients $\dfrac{d\Omega}{da}$, $\dfrac{d\Omega}{dy}$ being the partial ones in regard to a, y respectively. The equation $\Omega = 0$ signifies that the variation of x, in so far as it arises from the variation of the constants, is $= 0$; that is, the equation

$$\Delta x = \frac{dx}{da}da + d\beta = 0,$$

where α gives ν, y is the constants a, u, c, the differentiations $\dfrac{dx}{da}$, $\dfrac{dx}{db}$, &c. being the partial ones in regard to a, b, respectively. The differential $\Delta x = 0$ signifies that the solution of a, in so far as it arises from the variation of the constants, is $= 0$; (2) the derived derivatives of a in regard to the constants a, b, &c.

$$\frac{dx}{da}\,d\alpha + \frac{dx}{d\beta}\,d\beta + \frac{dx}{d\gamma}\,d\gamma + \frac{dx}{d\delta}\,d\delta = 0,$$

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The value of $x'\left(=\dfrac{dx}{dt}\right)$ is therefore obtained from that of x by differentiating in regard to t alone, as if a, c, e, ν were constants: viz. if x' is given as a function of t, a, c, e, ν, then the variation of x' in so far as it arises from the variation of the constants, $\Delta x' = \dfrac{dx'}{da}\,da + \dfrac{dx'}{dc}\,dc + \dfrac{dx'}{de}\,de + \dfrac{dx'}{d\nu}\,d\nu$ remains.

There are the like equations in regard to y, z; viz. in all, four equations which are linear in regard to $\dfrac{da}{da}, \dfrac{dc}{da}, \dfrac{de}{da}, \dfrac{d\nu}{da}$ and which serve to determine these quantities in terms of α, β, γ, δ.

Now considering the x, y as expressed in terms of a, c, e, $\ldots$, then Ω becomes a function of these quantities; the differential coefficients $\dfrac{d\Omega}{da}$, &c., being connected with the original differential coefficients $\dfrac{d\Omega}{dx}, \dfrac{d\Omega}{dy}$ by the equations

$$\frac{d\Omega}{da} = \frac{d\Omega}{dx}\cdot\frac{dx}{da} + \frac{d\Omega}{dy}\cdot\frac{dy}{da}, \quad \&c.$$

As there are four equations, $\dfrac{d\Omega}{dx}, \dfrac{d\Omega}{dy}$ can be expressed in an infinity of ways in terms of $\dfrac{d\Omega}{da}, \dfrac{d\Omega}{dc}$, &c., and, considering $\dfrac{d\Omega}{da}$, &c., as given in terms of $\dfrac{d\Omega}{dx}, \dfrac{d\Omega}{dy}$, we can in an infinity of ways express $\dfrac{da}{da}$, &c., as linear functions of $\dfrac{d\Omega}{da}, \dfrac{d\Omega}{dc}$, &c. But there is no form (obtained by combining the equations in a particular manner) wherein the coefficients of the non-constant quantities are functions of a, c, e, $\ldots$ without a contact; and this is the form most simply employed for the expression of $\dfrac{da}{da}$, &c. in terms of $\dfrac{d\Omega}{da}, \dfrac{d\Omega}{dc}$, &c. in the method wherein these quantities are made use of in this respect.

I remark upon the present question, that the answer ought to be in evidence perfectly familiar to every student in *Physical Astronomy*; and that a student ought to be able to present it in a clear and lucid form: the question being in fact intended as a test of ability in this respect.

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13. *Explain the course of the geodesic lines on a spheroid of revolution; and in particular show that the meridian is included in-virtue of which any geodesic line, com- sidered as starting from a given point, cannot at every passage of the vertex be a shortest line.*

From each point on the surface a geodesic line may be drawn in any direction whatever along the surface, that is, through each point of the surface there is a singly infinite series of geodesic lines. A geodesic line undulates (in the manner of a sinuous sinusoidal between two parallels of latitude, called the upper and lower parallels respectively; viz. considering it as starting from a point a on the upper parallel, it descends to the equator, then to the lower parallel, ascends to the equator, on the upper parallel and again, viz. this is between A, B, C, D, on the equator, on the upper parallel and again, viz. on the upper limit, C being the upper limit, B is the lower limit, and similarly the parallels between C and A, D and B, are the same so on. Now, in general, the equator is or upper and a lower at the same time; so that, in general, D does not concide the equator, A being given, A ascending will be: D, will be a portion of a shorter line.

The equatorial are AAA all A, the meridian portions of these the in- crease between as long D have 100°, the value of an become infinitely large 180°. If we are in question is commensurable with 100°, the geodesic line will be a closed curve; otherwise to its case, the limit is not, then it is not a closed curve, but presents oscillating for ever between the two parallels. It is the limiting case where the inclination is = 90°, the geodesic line is obviously a meridian.

Considering a geodesic line starting in a given direction from a point A, and the geodesic line from the same point A in the consecutive direction, it appears from the foregoing account of the configuration of the lines, that the two lines will inter- sect each other in general an infinite number of times: supposing that they first intersect in a point B, then by a general theorem of Jacobi's, the geodesic line AB is a shortest line from B to any point nearer than B; it is a shortest line so far as the geodesic line is obviously a meridian.